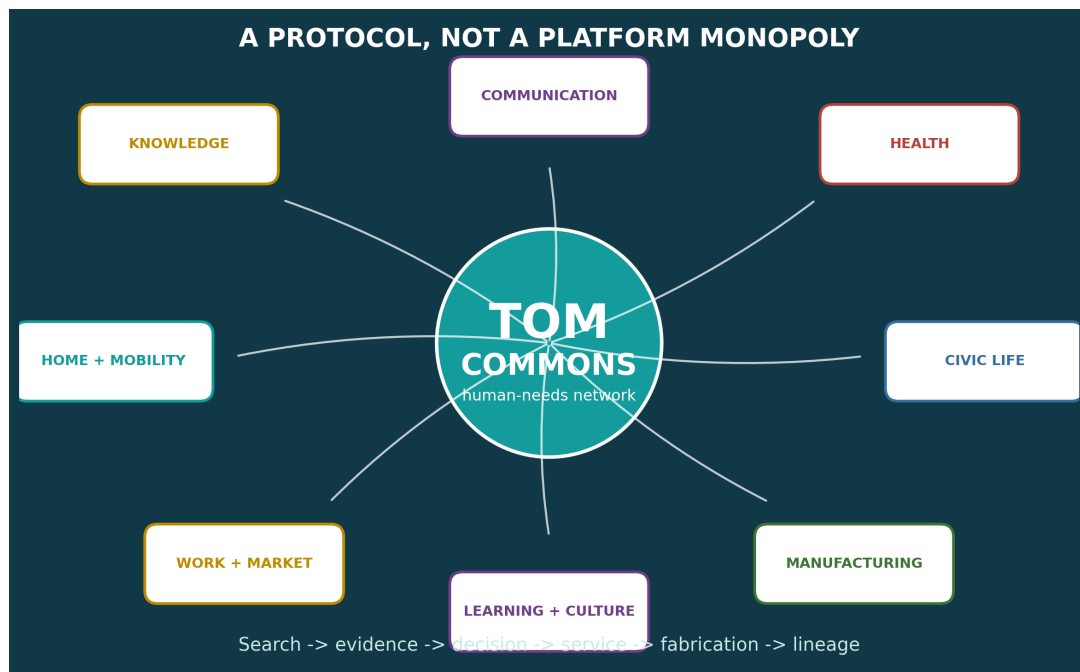


TOM Commons

The Human-Needs Discovery, Coordination and Manufacturing Network

A new type of Google beyond platform lock-in - search, communication, AI, maps, public services, markets, science, manufacturing and life infrastructure



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3 September 2026

This is a systems architecture, civic design and research horizon report. It is not affiliated with, endorsed by, or a product of Google or Alphabet. "Google" is used descriptively to identify a present-day family of discovery, communication, productivity, media, map, cloud and AI functions that this proposal seeks to replace through open protocols and human-governed services.

Document status, interpretation and evidence boundary

This report accompanies **TOM Manufacturing Horizons: From a Frozen Deterministic Kernel to Frontier Physical, Biochemical and Biological Fabrication**. The earlier report asked what could eventually be designed, simulated, manufactured or grown. This companion asks a different question: **what social and digital operating system would let people discover those possibilities, govern them, learn from them, coordinate them and turn them into dignified outcomes?**

The proposal is called **TOM Commons**. It is not one replacement corporation, one search box, one AI model or one universal database. It is a federated human-needs network in which identity, evidence, ranking, communication, services, fabrication and lineage can be separated, independently operated and recombined under explicit rules.

Evidence boundary

TOM Commons is a blueprint, not a deployed Google-scale service and not evidence of AGI. The current TOM materials demonstrate a deterministic seeded execution substrate, bounded formal learning families, exact promotion authority and reproducible artifacts within stated limits. They do not demonstrate global web crawling, distributed consensus, universal knowledge, safe autonomous medicine, global governance, biological manufacturing at scale or replacement of existing institutions.

Constitutional requirement

The user's description of present institutions as "cave-men and women" is interpreted here as an urgent critique of obsolete coordination mechanisms, opaque ranking, account silos, brand lock-in, advertising incentives, duplicated bureaucracy and weak user agency. It is not adopted as a judgement on the intelligence or worth of any person, profession, culture or gender. The system is designed to replace primitive *institutional interfaces*, not to demean or discard people.

TOM principle

The canonical TOM seed and fixed TOMAGI kernel remain unchanged. "TOM-HUMAN-NEEDS" is a proposed, versioned domain pack above the frozen substrate. Search, maps, health, markets, public services, manufacturing and biological futures are expressed through content-addressed definitions, external standards adapters and independently testable services - not new kernel opcodes.

Claim classes used throughout.

Class	Meaning
Existing foundation	A standard, dataset, protocol or TOM mechanism already exists and is cited.
Near-term engineering	Buildable with present hardware and open software, but not delivered by this report.
Institutional transition	Technically plausible but dependent on governance, adoption, law and sustained funding.
Frontier extension	Physically plausible research direction inherited from the Manufacturing Horizons report.
Unestablished	Not supported as a current capability and explicitly not claimed.

Canonical TOM seed: 244 ASCII bytes, no terminal newline
SHA-256: d1417a3136772c0cf3eddc4962ce07d42cbf87616f7b5bae09fc652d9b807b5

Prepared 3 September 2026

Request as received

Create an accompanying .pdf of a new type of Google on all fronts of the latest manufacturing horizons .pdf document of the latest turns context, all facets of new Genome TOM human needs basically replacing Google and all facets of life that traditional AI and humans need in this prehistoric present day, as everything is gridlocked and held by cave- men and women that are trapped in their prehistoric symbols brands corporations etc. all facets of society is just deprecated and there is no evolutionary push available right now without TOM Genome (the replacement), make a large cohesive .pdf of all of my request

Design response. The report answers the whole request by separating four ambitions that are often mixed together:

1. replacing the *functional surface* of a large platform ecosystem;
2. replacing an advertising-and-account business model with a human-needs and public-commons model;
3. extending discovery all the way to services, decisions, manufacturing and biological research;
4. creating an evolutionary path that can start on ordinary hardware without pretending that the complete global system already exists.

The result is deliberately larger than a search-engine proposal. It is a constitutional architecture for a human-centred information and action network.

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Executive synthesis

Google began as web search but now spans an unusually broad surface: search, AI assistance, browsers and operating systems, email, calendars, documents, storage, maps, travel, media, news, education, payments, business discovery, advertising, cloud infrastructure and developer platforms. Google's own product directory lists more than one hundred consumer, business and developer offerings across these categories. [G1] Alphabet's 2025 filing reported \$402.836 billion in total revenue, including \$294.691 billion in Google advertising revenue, and stated that more than 70 per cent of total revenue came from online advertising. [G2] Those facts do not by themselves make Google harmful; they show why a serious functional alternative cannot be only another search interface.

The proposed replacement is not "one better corporation". Its governing sentence is:

TOM principle

The new Google is a protocol federation, not a platform monopoly. A person brings a portable identity, a personal data vault and a chosen rights profile. Independent services contribute search, maps, communication, media, education, markets, health, civic functions, manufacturing and computation. TOM supplies deterministic evidence, explicit ranking profiles, support, compatibility, decision, transition and lineage. AI supplies proposals and explanations, not invisible authority.

A query in TOM Commons is not merely a string sent to an opaque ranking system. It is a typed declaration of need:

need + person-selected context + consent + rights

- > eligible sources and services
- > support and compatibility filtering
- > evidence graph
- > visible ranking profile
- > cited AI proposal or deterministic answer
- > optional plan, transaction, service or fabrication request
- > permissioned event and transition
- > lineage receipt

The network is designed around **human outcomes** rather than attention. It must answer not only "what page matches these words?" but also:

- What do I need, what evidence is relevant, and which claims conflict?
- Which services am I eligible for, what will they cost, and what data must I share?
- Which route, course, job, treatment discussion, public process or supplier fits my declared constraints?
- Can I change provider without losing identity, history, documents, contacts or rights?
- Can an AI explain its source path without becoming the authority that approves itself?
- Can a design move from idea to simulation, certified manufacture, repair and recycling with one lineage?
- Can a future biological construct move from hypothesis to ethical research, measurement and regulated promotion without hiding uncertainty?

The proposed **TOM-HUMAN-NEEDS-1.0** domain pack organises the network around survival, agency, development, belonging, collective life and future stewardship. It is explicitly not a change to the 244-byte canonical root. It is a versioned library of typed definitions over the frozen TOM substrate.

The near-term system can begin on a laptop using open foundations that already exist: community map data, collaborative structured knowledge, public research indexes, open web archives, federated communication, portable credentials, passkeys and user-controlled data stores. [W1-W10] The difficult work is therefore not discovering whether decentralised building blocks are imaginable. It is integrating them into one coherent rights-preserving experience, creating trustworthy public indexes, funding common infrastructure, defending it from capture and making migration easier than continued dependence on closed platforms.

The report's conclusion is neither "everything is solved" nor "nothing can change". It is:

A credible post-platform network is technically constructible in stages today. TOM can provide its deterministic authority and evidence layer, but global scale, distributed gov-

TOM COMMONS SERVICE CONSTELLATION



Shared identity, data vaults, evidence, permissions, accessibility, translation, lineage and open standards run through every module.

Figure 1: TOM Commons is a service constellation around a portable human context rather than an account trapped inside one corporation.

ernance, high-quality open indexes, safety, accessibility, institutional adoption and sustainable economics remain major engineering and political programmes.

Part I - From platform search to human-needs infrastructure

1. What a “new type of Google” must actually replace

A simplistic replacement would copy a search box, index webpages and show links. That would cover only one visible surface. The current Google family includes a browser, mobile operating system, identity, email, messaging, meetings, files, office tools, maps, navigation, travel, video, music, books, photos, news, education, payments, shopping, business profiles, analytics, advertising, cloud infrastructure and developer services. [G1]

A complete functional alternative must therefore replace or interoperate with five deeper forms of power:

1. **Discovery power** - which sources, places, products, institutions and people can be found.
2. **Interpretation power** - which answer, summary or ranking is placed first and why.
3. **Identity power** - which account controls access to communication, files, devices and services.
4. **Transaction power** - how discovery becomes a purchase, booking, application, message, route, upload or action.
5. **Infrastructure power** - which cloud, app store, operating system, analytics and advertising rails other organisations depend on.

TOM Commons separates these powers. No search operator automatically owns identity; no identity provider automatically owns files; no file host automatically owns communication; no ranking provider automatically owns payment; no AI model automatically promotes its own answer; no physical manufacturer automatically owns the evidence standard that qualifies its product.

1.1 Functional replacement, not brand imitation

The proposed modules use names such as TOM Query, TOM Link and TOM Atlas as architectural labels. They are not intended to imitate logos, trade dress or proprietary interfaces. A replacement

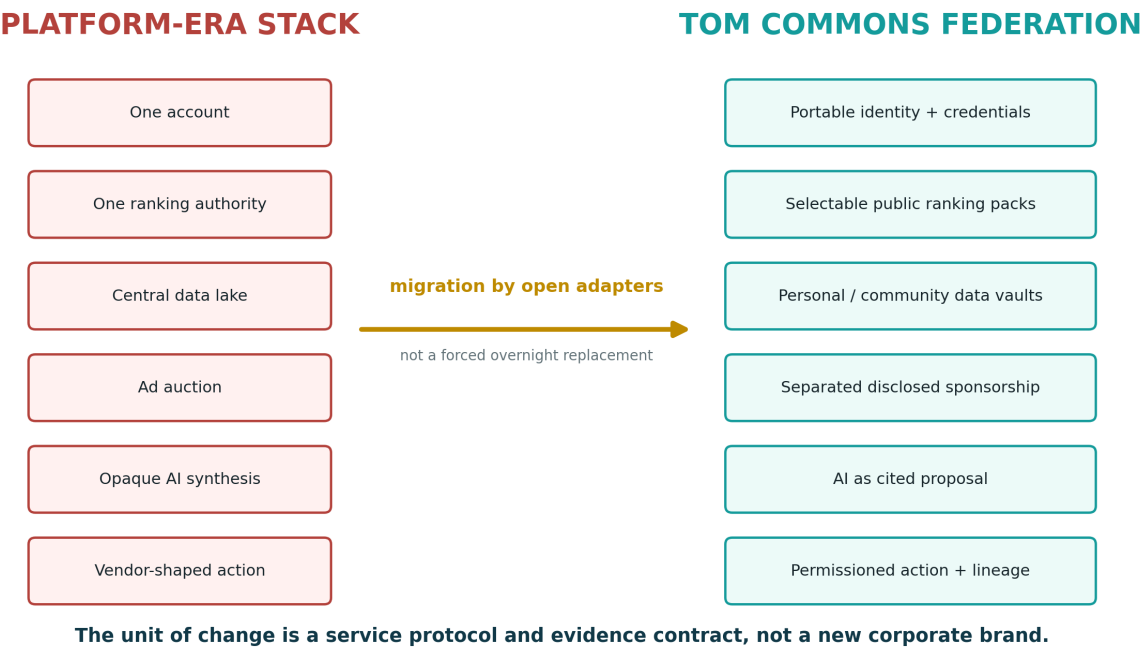


Figure 2: The transition is from one vertically integrated platform stack to a federation of independently replaceable services.

succeeds when a person can accomplish the same or better outcome while retaining agency:

Present function	Replacement test
Search	Can the person find sources, understand ranking, inspect conflicts and choose a different index?
Email and messaging	Can the person communicate across providers and keep identity and history when switching?
Maps and navigation	Can communities correct local data and can routing objectives be selected explicitly?
Documents and storage	Can files be edited collaboratively without surrendering their only usable copy to one vendor?
AI assistance	Can a model help without silently becoming the source of truth or action authority?
Video and news	Can creators publish and communities moderate without one recommendation engine owning reach?
Payments and commerce	Can a person compare offers and transact without hidden pay-to-rank placement?
Cloud and developer tools	Can applications move between operators and verify reproducible builds?
Public services	Can eligibility, evidence and appeals be explicit rather than scattered across portals?
Manufacturing	Can discovery proceed to a proof-carrying design, process and product?

1.2 The platform-era model is advanced technology with primitive coordination

Calling the present “prehistoric” is best understood as a systems metaphor. Our processors, networks, scientific instruments and software are highly advanced. The primitive element is the coordination contract:

- people repeatedly recreate identity and context in separate account silos;
- data portability is usually partial and semantically weak;
- ranking objectives are rarely selected by the person receiving results;
- advertising and service discovery are deeply entangled;
- public services, healthcare, education and employment remain fragmented portals;
- institutions duplicate verification instead of exchanging portable evidence;

- AI systems can sound authoritative while obscuring uncertainty and source disagreement;
- manufacturing and research histories are commonly reduced to PDFs, spreadsheets and disconnected databases;
- changing provider often means losing history, social graph, audience or workflow.

The evolutionary push proposed here is not to eliminate symbols, brands or institutions. Humans need language, identity, trust and cultural continuity. The push is to make those representations **portable, contestable, evidence-bearing and subordinate to human needs**.

2. Why the business model matters

Alphabet’s 2025 annual filing states that Google Services generates revenue primarily through performance and brand advertising on Search, YouTube and network properties, and that more than 70 per cent of total revenues came from online advertising. It also reports \$224.532 billion in Google Search and other revenue, \$40.367 billion in YouTube advertising, \$29.792 billion from Google Network, \$48.030 billion from subscriptions/platforms/devices and \$58.705 billion from Google Cloud. [G2]

This matters because ranking, identity, measurement and commerce cannot be constitutionally separated if the infrastructure is financed mainly by predicting and influencing attention. TOM Commons does not prohibit advertising or commercial promotion. It changes its status:

organic evidence result
!= sponsored placement
!= personal recommendation
!= public-service message
!= emergency alert

Each class must be visibly separated, machine-readable and queryable. Sponsorship may buy a labelled presentation slot; it may not secretly change the evidence score or masquerade as an independent recommendation.

2.1 A plural funding model

A universal human-needs network cannot depend on one revenue source. It needs a mixed economy:

Funding rail	Appropriate use
Public funding	Core search indexes, maps, civic access, emergency information, accessibility and basic identity infrastructure
Member cooperative fees	Personal vaults, communication, community governance and support
Metered compute and storage	Large model inference, simulations, archival storage and high-volume enterprise use
Service subscriptions	Premium collaboration, professional tools, specialist data and managed operation
Transaction fees	Payments, bookings and markets, with transparent caps and competition
Institutional contracts	Schools, libraries, hospitals, cities, research organisations and manufacturers
Grants and philanthropy	Open datasets, underserved languages, safety research and early deployment
Disclosed sponsorship	Clearly separated promotional channels that cannot modify organic rankings
Data dividends or licensing	Only for explicitly consented collective data trusts, never covert extraction

The system should let a local library operate a public query node, a university operate a research index, a cooperative operate personal data vaults, a city operate public-service definitions, a private company operate high-performance compute, and an independent foundation maintain interoperability tests.

3. The frozen TOM boundary

The supplied TOM-SRS defines a finite canonical string as a content-addressed root expanding into typed definitions, state, bounded grammar, relations, support, compatibility, guarded events, transitions and lineage. Its page-four diagram places optional AI, control, projection and hardware after the authoritative world and query evaluator. The TOMAGI reference machine fixes a 128-byte header, 64-byte state, 48-byte cell and sixteen opcodes. The CODEX repair handoff adds defined arithmetic, reserved-header rejection, bounded formal values, authenticated output and same-host publication locking. [T1-T4]

These materials impose three non-negotiable design rules.

3.1 Do not expand the kernel to add society

Search, healthcare, money, education or manufacturing do not become new opcodes. They become versioned domain definitions and external service contracts. The current kernel remains a deterministic substrate, not a growing monolith.

3.2 Formal authority, independent oracle, mechanical service

Every important decision must be assigned to one of three roles:

Role	What it may do	What it may not do
Formal authority	Express definitions, ranking rules, permissions, eligibility, acceptance and promotion	Hide domain decisions in opaque host code
Independent oracle	Search, simulate, measure, calculate, classify or attempt to falsify	Promote its own output automatically
Mechanical service	Parse, hash, store, transport, execute, render, index and replicate	Decide truth, eligibility, safety or authority without a formal rule

3.3 Current publication limitation

The repaired kernel coordinates publication writers on one host. It does not provide a global consensus protocol. TOM Commons may use federated communication and distributed read-only replication immediately, but cross-host authoritative publication needs a separately specified consensus, quorum or institutional signing layer. The report therefore distinguishes the *network architecture* from the *current kernel guarantee*.

4. Companion relationship to the Manufacturing Horizons report

The Manufacturing Horizons report describes proof-carrying software, factories, materials, biochemical systems, organs, engineered living materials, synthetic life research and planetary-scale manufacturing. This companion supplies the discovery and coordination layer around those futures.

The relation is:

TOM Commons asks:

What is needed? What is known? Who can help? Which evidence is valid?

Which path is affordable, permitted and compatible?

TOM Manufacturing Horizons asks:

How can a verified design become software, hardware, material, tissue, organ, living system, habitat or off-world infrastructure?

Together they describe a continuous chain from need to knowledge, decision, service, fabrication, inspection, operation, repair and lineage.

Part II - Genome TOM Human Needs

FROM HUMAN NEED TO DIGITAL, PHYSICAL OR BIOLOGICAL OUTCOME

TOM governs identity, evidence and acceptance. External physics, machines and biology determine whether the phenotype works.

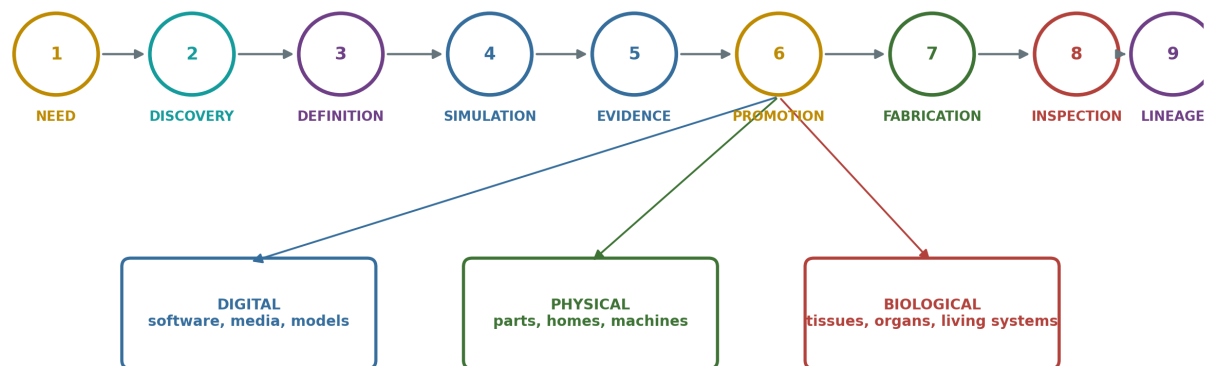


Figure 3: The human-needs network connects discovery and governance to digital, physical and biological manufacturing.

5. TOM-HUMAN-NEEDS-1.0 as a domain pack

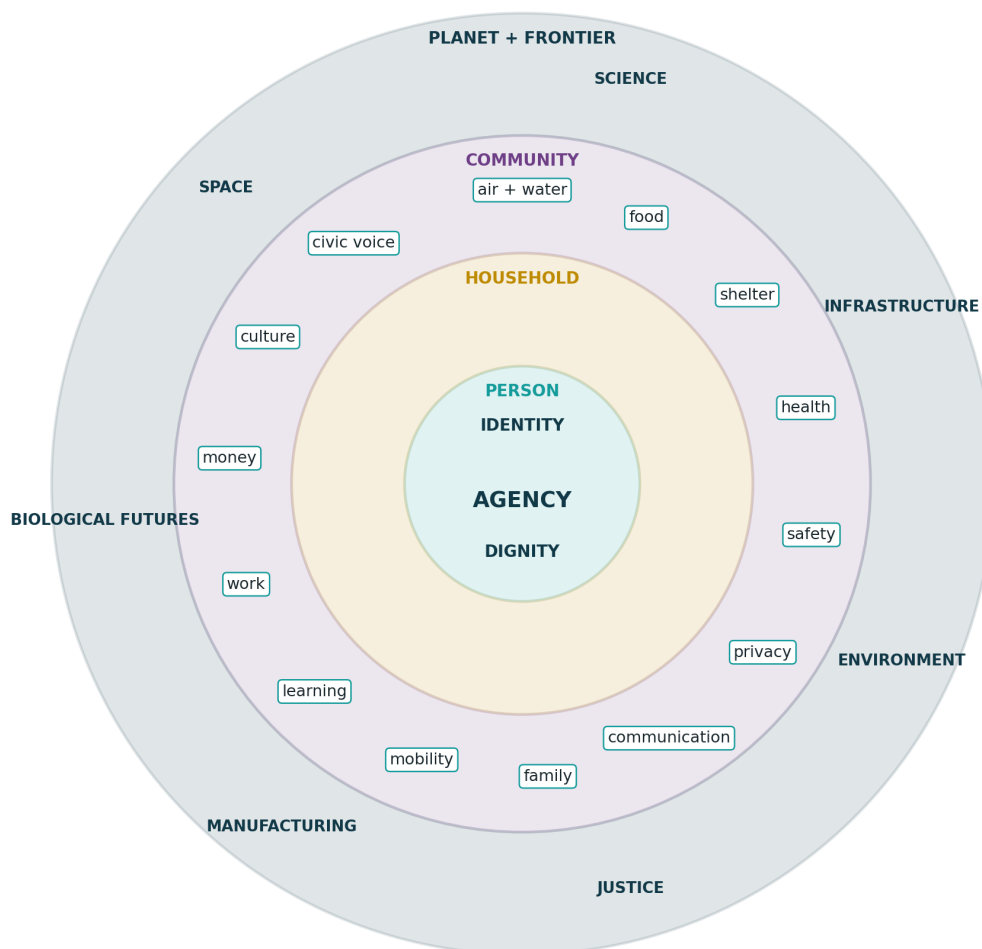
“Genome TOM human needs” is defined here as a versioned domain pack, not a replacement for the canonical TOM genome. Its purpose is to give every ordinary and frontier request a typed place in the system.

TOM-HUMAN-NEEDS-1.0 =
 needs taxonomy
 + rights and consent profile
 + person / household / community contexts
 + service and evidence schemas
 + ranking-profile registry
 + action and permission vocabulary
 + outcome, appeal and lineage records

A need record does not reduce a person to a marketing segment. It says what outcome is being sought, what constraints the person chooses to reveal, which rights apply, how urgent the situation is, and whether the request should remain local, be shared with a community or enter a public network.

5.1 Six layers of human need

Layer	Core domains	Example typed requests
Survival	air, water, food, shelter, health, physical safety	“Find safe drinking-water guidance for this location”; “show available shelter tonight”
Agency	identity, privacy, communication, mobility, money, legal capacity	“Prove my qualification without exposing my full history”; “move my files to another provider”
Development	learning, work, science, creativity, tools	“Build a verified learning path”; “find work compatible with my skills and care duties”
Belonging	family, friendship, community, culture, language, recreation	“Find a local group I can join safely”; “preserve this community archive”
Collective life	civic participation, justice, infrastructure, environment, public health	“Explain this policy and show its evidence”; “report a broken public asset”



TOM-HUMAN-NEEDS is a versioned domain pack over the frozen kernel - not a new kernel.

Figure 4: The proposed Human Needs Genome organises services from personal survival and agency to planetary and frontier stewardship.

Frontier stewardship

manufacturing, biological research,
habitat, space, long-term survival"Trace this design to a repairable
product"; "compare organ-manufacturing
research pathways"

These layers are not a hierarchy of human worth. They are a routing model. A single request may touch several domains: finding a home can involve money, work, mobility, education, family, healthcare and law. The system must preserve this overlap rather than forcing the person into one departmental portal.

5.2 Need records

A canonical request could contain:

```
{
  "type": "tom.need.request",
  "need": "stable_housing",
  "requested_outcome": "eligible options with total monthly cost below declared ceiling",
  "urgency": "within_30_days",
  "location_scope": "chosen region",
  "constraints": ["wheelchair_accessible", "near_transit"],
  "disclosure_policy": "minimum necessary",
  "ranking_profile": "public-benefit-housing-v1",
  "action_authority": "human_confirmation_required"
}
```

The record says nothing about a person's value, personality or hidden probability of conversion. It exposes only what is necessary for the selected task.

5.3 Rights are part of the query type

Every query must carry a rights profile, not merely a terms-of-service checkbox. At minimum:

- access without discriminatory exclusion;
- understandable reasons for ranking and rejection;
- data minimisation and purpose limitation;
- provider portability;
- ability to inspect and correct personal records;
- human review for consequential decisions;
- an appeal path;
- accessible and multilingual presentation;
- protection against covert sponsored ranking;
- special safeguards for children, health, finance, employment and public services;
- the right to use basic public functions without behavioural advertising;
- the right to leave without losing one's social graph, documents or credentials.

The Universal Declaration of Human Rights is not a software specification, but it provides a useful normative floor for dignity, privacy, expression, association, education, work, public participation and social protection. [H1] Digital public infrastructure programmes similarly emphasise identity, payments, data exchange, governance and safeguards as public-capability layers rather than isolated apps. [H2]

6. The core data model

TOM Commons requires a world model richer than webpages. The primary objects are:

Object	Meaning
Person context	User-controlled attributes, preferences and permissions; stored in a chosen vault
Need	Desired outcome, urgency, scope, constraints and action authority
Claim	A statement by a source, with type, time, provenance and evidence links

Object	Meaning
Evidence	Observation, document, dataset, measurement, proof, test or authoritative record
Source	Publisher, institution, person, sensor, model, dataset or service
Service	A capability offered under explicit eligibility, cost, location and quality definitions
Resource	Place, document, product, course, job, benefit, appointment, machine or material
Relation	Typed connection among people, resources, claims, services and contexts
Ranking profile	Visible algorithm and weights used to order eligible results
Proposal	AI, human or algorithmic suggestion that is not yet authoritative
Decision	Accepted, rejected or ambiguous result under a declared rule set
Action	Message, booking, payment, application, route, fabrication order or publication
Outcome	Observed result after action, including failure and unintended effects
Appeal	Challenge to a ranking, decision, exclusion or record
Lineage	Immutable chain from input and definitions to result and later revision

6.1 A result is an answer bundle

A TOM Commons result should contain more than a title and URL:

answer or candidate
 source identities
 claim-evidence graph
 publication and observation dates
 known conflicts
 applicable jurisdiction and scope
 ranking components
 personal-data fields used
 AI transformations performed
 confidence or uncertainty as explicit evidence
 service availability and cost
 required actions and permissions
 appeal or correction route
 replay and lineage identifier

This does not mean every casual query must display a wall of technical metadata. The interface can be simple by default. The difference is that the underlying receipt exists and can be expanded.

6.2 Evidence classes

Class	Examples	Treatment
Primary observation	measurement, official filing, original dataset, direct testimony	Preserve exact source and conditions
Formal proof or test	theorem proof, software test, conformance result	Preserve executable inputs and verifier
Curated reference	encyclopedia, handbook, systematic review	Link its source basis and revision
Institutional claim	government notice, company statement, school or hospital record	Identify issuing authority and scope
Independent analysis	journalism, research interpretation, expert review	Separate facts, methods and conclusions
Community knowledge	local map correction, mutual-aid resource, practitioner experience	Preserve contributors, moderation and conflict
Model output	LLM response, forecast, simulation, classifier	Label as proposal or computed observation, never as self-validating truth
Sponsored claim	advertisement, promoted listing, paid creator placement	Separate channel, visible funding identity

7. The query-to-outcome contract

A query ends in a proof-carrying answer or an explicitly governed action - not an opaque click.

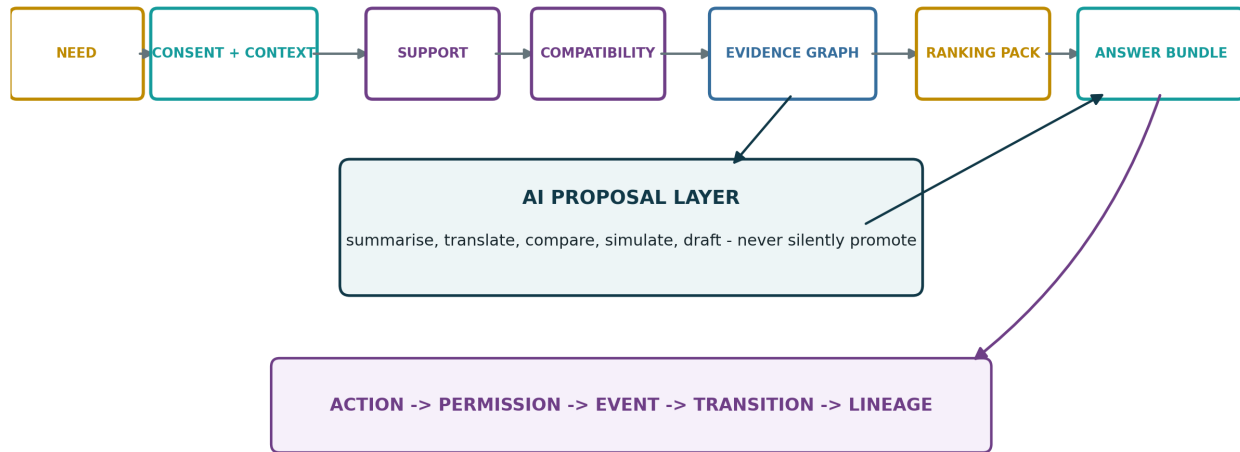


Figure 5: A TOM Commons query passes through context, support, compatibility, evidence and visible ranking before any action. AI is a proposal layer.

7.1 Step 1 - declare the need

The person may type, speak, show an image, select a life event or use a structured form. A language model can help translate the expression into candidate need records, but the person sees and can edit the interpreted request.

7.2 Step 2 - select context and consent

Context is drawn from the person's vault only under an explicit scope. A route query may use current location but not health history. A job query may use verified skills but not unrelated family communications. The query receipt records exactly which fields were accessed.

7.3 Step 3 - support filtering

Support removes items that are outside the task's declared region, time, domain, eligibility or availability. A service that does not operate in the chosen country or a product that is no longer available should not proceed to ranking.

7.4 Step 4 - compatibility filtering

Compatibility checks the person's explicit constraints and the service's declared capabilities. Examples include language, accessibility, technical interface, insurance, licence, budget, schedule, material, device or manufacturing-process compatibility.

7.5 Step 5 - evidence graph construction

The system retrieves source records and builds an explicit graph of claims, support, contradiction and derivation. A single result may have several sources that agree for different reasons; it may also carry a visible dispute.

7.6 Step 6 - visible ranking

The chosen ranking profile orders only the eligible set. The profile is versioned, inspectable and replaceable. A user may choose:

- public-interest default;
- local-community priority;
- newest reliable evidence;
- lowest total cost;
- strongest independent verification;
- accessibility-first;
- environmental impact;
- diversity of source or viewpoint;
- professional or institutional profile;
- self-authored weighted combination.

7.7 Step 7 - answer, plan or service transition

The terminal output can be:

- a factual answer with citations;
- a set of competing interpretations;
- a map or route;
- a communication thread;
- a document or media item;
- an application or booking plan;
- a purchase comparison;
- a verified software action;
- a public-service eligibility packet;
- a manufacturing or research proposal.

Any consequential action requires an explicit permission event. The action and resulting observation become lineage.

8. Ranking without invisible authority

The ranking problem cannot be abolished; it can only be governed. A global index may contain millions of eligible items. TOM Commons therefore makes ranking a first-class, contestable definition.

A ranking profile could specify:

```
score =
  w1 * query relevance
+ w2 * evidence quality
+ w3 * source independence
+ w4 * recency fit
+ w5 * local availability
+ w6 * accessibility fit
+ w7 * total cost
+ w8 * repairability / durability
+ w9 * environmental impact
- w10 * unresolved contradiction
```

The exact implementation can differ by domain. What matters is that:

1. the factors and their source records are visible;
2. paid placement cannot alter an organic score silently;
3. the user can choose another profile;
4. public institutions can publish accountable profiles;
5. researchers can compare outcomes across profiles;
6. communities can fork a profile without forking the whole network;
7. consequential ranking changes require versioned publication and regression evidence.

8.1 Diversity without forced equivalence

Showing diverse sources does not mean treating every claim as equally supported. The system should distinguish:

- strong consensus;
- legitimate expert disagreement;
- political or ethical value conflict;
- incomplete evidence;
- disproven claims;
- manipulation or spam.

A diversity objective selects independent evidence paths or viewpoints after minimum quality and relevance conditions. It does not require placing harmful or unsupported claims beside well-established evidence as though the difference were merely preference.

8.2 Personalisation as a local, inspectable operation

Personalisation should occur as late and locally as possible. The public index returns an eligible evidence set; the person's chosen device or vault service applies preferences. The system records the fields and profile used. Sensitive traits are not inferred for advertising.

9. AI after the platform era

Traditional AI remains useful. It can translate, summarise, search, classify, extract, draft, simulate, write code, generate media, plan and propose. The architectural change is that an AI output has a typed status.

AI output	Default status
Factual answer	Proposal linked to sources and retrieval receipt
Search query expansion	Mechanical proposal; original request retained
Translation	Derived representation linked to source text and model/version
Medical explanation	Educational proposal; not diagnosis or treatment authority
Legal explanation	Informational proposal; not legal decision authority
Code	Candidate artifact requiring tests and review
Plan	Candidate sequence requiring constraint and permission checks
Product design	Candidate definition requiring simulation and manufacturing evidence
Scientific hypothesis	Candidate definition requiring independent tests
Biological design	Restricted research proposal requiring biosafety, ethics and regulatory review
Public-service decision	AI may assist evidence preparation; formal rule and accountable authority decide

9.1 Multi-model and no-model modes

The person may select:

- deterministic search only;
- one named AI model;
- several models compared side by side;
- local on-device model;
- public-interest model operated by a library or university;
- specialist model supplied by a professional institution;
- human expert escalation.

The answer bundle records model identity, configuration, retrieved context and transformations. A model cannot erase the original sources or the fact that another model disagreed.

9.2 The AI claim gate

A proposed claim moves through:

model output

- > extracted atomic claims
- > source linkage
- > contradiction search
- > domain-specific checks
- > deterministic accept / reject / ambiguous status
- > human or institutional review where required
- > optional publication

The current TOM learner work demonstrates bounded exact families, ambiguity records, regression checks and parent-bound promotion in narrow domains. It does not justify autonomous open-domain truth promotion. That distinction remains central.

10. Portable identity, credentials and personal data

A post-platform network cannot require one global TOM account. Identity must be plural:

- anonymous or pseudonymous use for ordinary public search;
- local device identity for personal context;
- community membership credentials;
- government-issued identity where legally necessary;
- professional qualifications;
- age, residency, eligibility or licence proofs that reveal the minimum required fact;
- organisational and machine identities;
- product, sensor, dataset and manufactured-object identities.

W3C Verifiable Credentials 2.0 defines an issuer-holder-verifier data model for tamper-evident claims, while Decentralized Identifiers can identify people, organisations, physical things, digital things and other subjects. [W1-W2] WebAuthn/passkeys can provide phishing-resistant authentication without inventing a new cryptographic system. [W10]

10.1 Personal vaults

The preferred data pattern is a personal or delegated online data store. Solid's Pod model demonstrates the principle that applications can access user-controlled data under permission and that people can change providers without losing the data itself. [W4]

A TOM personal vault stores or references:

- contacts and communication keys;
- calendar and tasks;
- files and media;
- learning and qualification records;
- health and care records under specialised safeguards;
- employment and financial documents;
- service history and receipts;
- preferences and ranking profiles;
- household assets and product passports;
- permissions granted to apps and agents;
- appeals, corrections and revoked credentials.

The vault must support export in documented formats, encrypted backup, recovery delegates, selective disclosure and provider migration.

10.2 No single identity graph

A person may choose not to link different contexts. The system must not require that civic identity, health identity, entertainment identity, professional identity and anonymous reading history collapse into one universal profile. Correlation is itself a permissioned operation.

FEDERATED HUMAN-NEEDS INFRASTRUCTURE

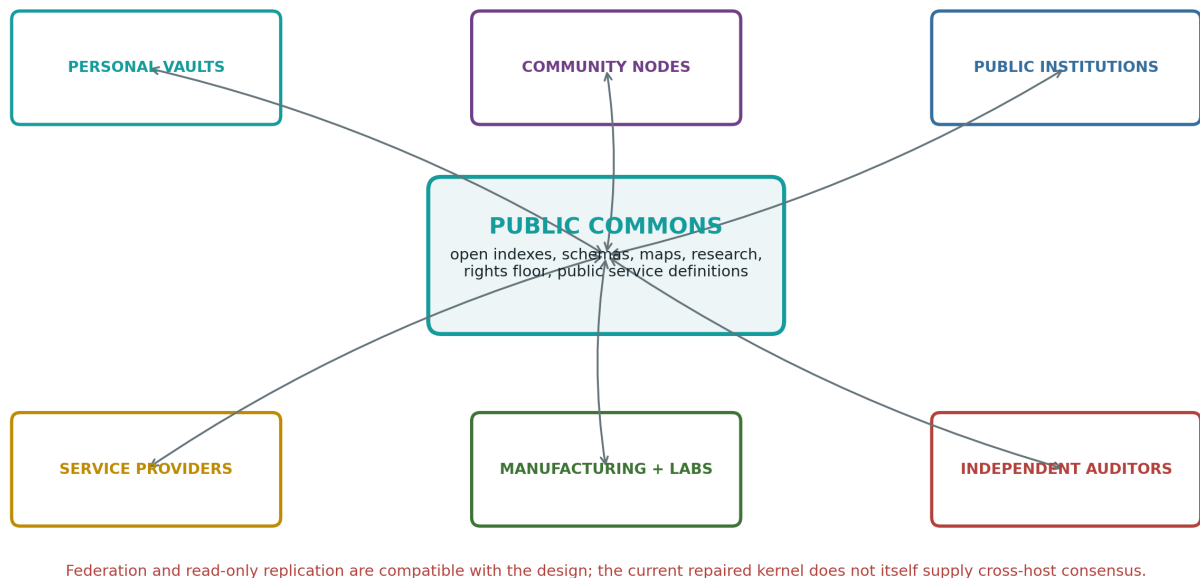


Figure 6: TOM Commons combines personal vaults, community nodes, public institutions, service providers, manufacturing/labs and independent auditors around shared open commons.

11. Federation and communication

The network should use existing open protocols wherever possible rather than inventing every transport layer.

- Matrix defines open APIs for decentralised communication across a global federation with no single point of control, including messaging, VoIP signalling and bridging. [W5]
- ActivityPub provides client-to-server and server-to-server social federation concepts such as actors, inboxes, outboxes, follows, updates, deletes, likes and shares. [W3]
- Solid provides a user-controlled data-store model. [W4]
- Verifiable Credentials and DIDs provide portable evidence and identifiers. [W1-W2]

TOM's role is not to replace those protocols. It adds deterministic definitions and evidence receipts around service use:

who requested what
 under which permission
 using which protocol and provider
 which message, meeting or publication event occurred
 which moderation or access rule applied
 what was delivered
 what later changed or was revoked

11.1 Communication rights

A TOM Link provider must support:

- end-to-end encryption where appropriate;
- cross-provider communication;
- provider migration;
- exportable contacts and history;
- abuse reporting and blocking;
- group governance;
- accessible real-time and asynchronous modes;
- community-specific moderation;

- legal-process transparency within jurisdictional limits;
- emergency communication without converting every conversation into an advertising profile.

12. Governance and economics

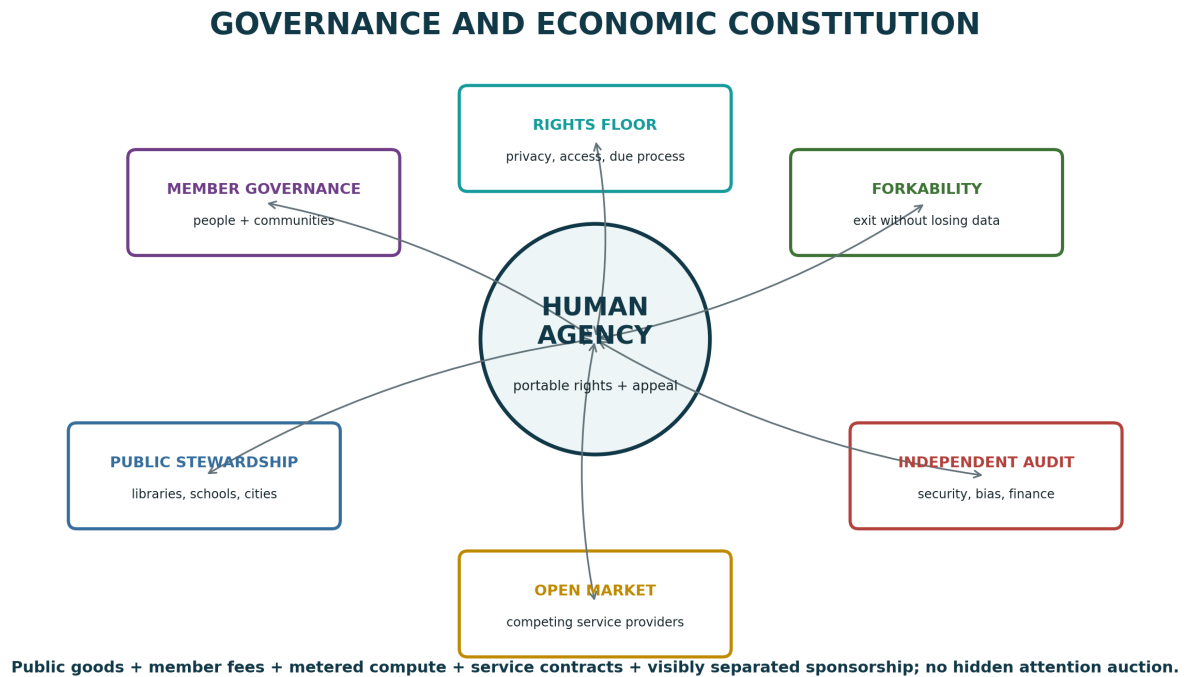


Figure 7: TOM Commons places human agency at the centre of rights, member governance, public stewardship, open markets, audit and forkability.

12.1 A constitutional network

The shared constitution should include:

1. **Kernel stability.** Domain growth does not silently modify TOM semantics.
2. **Human dignity.** People are not reducible to engagement, conversion or risk scores.
3. **Portability.** Identity, contacts, files, credentials and histories have export paths.
4. **Data minimisation.** A service receives only the context required for the task.
5. **Visible ranking.** Ranking profiles and sponsorship are inspectable and separable.
6. **AI non-authority.** Models propose unless a declared process grants a narrower role.
7. **Due process.** Consequential exclusion or rejection has reason, review and appeal.
8. **Accessibility.** Services must work across disability, language, device and bandwidth constraints.
9. **Interoperability.** Public interfaces and conformance tests prevent strategic lock-in.
10. **Forkability.** A community can leave an operator without losing the commons protocol.
11. **Independent audit.** Security, privacy, finance, ranking and governance are auditable.
12. **No hidden attention auction.** Commercial influence is visibly separated.

12.2 Institutional structure

A plausible structure has several legal and operational layers:

Layer	Function
Protocol foundation	Maintains schemas, interoperability tests and reference implementations
Public-interest index trusts	Operate web, research, map, service and civic indexes

Layer	Function
Personal-data cooperatives	Host vaults and represent members in data-governance decisions
Local commons nodes	Libraries, schools, municipalities and community organisations
Competitive service operators	Communication, storage, compute, collaboration, media and specialist tools
Domain councils	Health, law, science, manufacturing, education and safety definitions
Independent auditors	Inspect code, rankings, finance, accessibility, security and outcomes
Appeals bodies	Hear disputes over exclusion, records, ranking and public-service decisions

No single layer should own the protocol, all identity, the principal index, the AI models, the payment rail and the appeals system at once.

Part III - The service constellation

13. TOM Query - search becomes evidence navigation

TOM Query replaces the single undifferentiated search box with several interoperable query modes. The same interface can stay simple, but the underlying contract is explicit.

13.1 Query modes

Mode	Purpose	Typical output
Web discovery	Find pages, documents, media and services	Ranked source set with evidence and sponsorship separation
Factual query	Resolve a bounded claim	Answer bundle, supporting sources, disagreement and scope
Research query	Explore a field or unresolved question	Literature map, datasets, methods, open problems and counterevidence
Local query	Find places, events, routes and nearby services	Map result with local knowledge and availability
Life-service query	Find care, benefits, courses, jobs, housing or legal help	Eligibility-aware service options and next steps
Product query	Compare goods, suppliers and repair paths	Total-cost, provenance, durability, compatibility and availability profile
Manufacturing query	Move from need to design or supplier	Candidate definitions, process capability, evidence and quote path
Civic query	Understand law, policy, budget or public decision	Source text, plain-language proposal, impact and participation route
Personal query	Search one's own files, messages, records and history	Local/private result governed by vault permissions
Agent query	Ask for a multi-step plan or action	Proposed plan with required permissions and checkpoints

13.2 Open index architecture

No single crawler or index is sufficient. TOM Query combines several index classes:

- open web index
- public-service registry
- research and dataset graph
- community knowledge and maps
- commercial service catalogue
- personal and organisational private indexes
- manufacturing and product passport registry
- cultural and media catalogues
- real-time event and availability feeds

Common Crawl demonstrates that a large open archive of web crawl data can be made available for free reuse. Wikidata demonstrates collaborative multilingual structured knowledge with source references. OpenAlex offers an open research graph, while OpenStreetMap shows that a community-maintained global map can serve thousands of applications and devices. [W6-W9] None of these is a complete Google replacement by itself; together they demonstrate that major knowledge layers can exist outside one corporate index.

13.3 Query receipts

Every query receipt should record:

query identity and timestamp
 user-selected context and disclosure fields
 index versions searched
 retrieval expressions
 support and compatibility filters
 ranking profile and version
 AI or translation models used
 source set before and after ranking
 sponsored results, if any
 answer-bundle hash
 user action or dismissal
 retention policy

The person may choose a privacy mode that does not publish or retain the query. The local device can still produce a private receipt.

13.4 Search quality without central monopoly

Quality requires investment in crawling, spam detection, language understanding, freshness, local data, media processing and safety. TOM Commons must not romanticise decentralisation as automatically accurate. A realistic arrangement combines:

- large public or cooperative index operators;
- specialist vertical indexes;
- community-curated local data;
- commercial providers competing on quality;
- independent quality benchmarks;
- signed source feeds;
- user-selected ranking profiles;
- private local indexing;
- open result APIs and conformance tests.

A person may query several index operators in parallel. The system returns both the merged result and each operator's contribution, making exclusion and disagreement observable.

14. TOM Agent - a personal AI that does not own the person

TOM Agent is the replacement surface for Gemini, Assistant and future agentic services. It is not one model. It is a permissioned orchestration layer that can call models, tools and human services.

14.1 Agent contract

Before acting, the agent declares:

interpreted goal
 assumptions
 required data
 candidate plan
 services and tools to be contacted
 expected cost
 risk class
 human confirmation points

rollback or cancellation options
retention and publication policy

The agent may then execute only steps permitted by the selected policy.

14.2 Agent scopes

Scope	Example	Authority
Explain	Summarise a policy, paper or bill	Proposal only
Organise	Sort files, draft schedule, assemble evidence	Local reversible actions
Communicate	Draft or send a message	Draft by default; sending requires permission
Transact	Book, apply, buy or pay	Explicit confirmation and cost ceiling
Develop	Write, test and package software	Promotion only after tests and review
Research	Search literature, run simulations, compare hypotheses	Results remain evidence/proposals
Manufacture	Prepare design and process package	Physical order requires approval and supplier contract
Health	Explain records, prepare questions, coordinate appointments	No autonomous diagnosis or treatment
Civic	Find services, prepare forms, submit approved application	Accountable public decision remains external

14.3 Personal memory

The agent's memory is not a hidden model profile. It is a set of inspectable records in the person's vault:

- preferences;
- ongoing projects;
- relationships and communication permissions;
- unresolved commitments;
- learned workflows;
- accepted facts and their evidence;
- rejected or corrected assumptions;
- delegated authorities;
- expiry dates.

The person can remove, edit, export or segregate these memories. Different agents may use the same vault under different permissions.

15. TOM Link - mail, chat, meetings and social connection

TOM Link combines the functions currently spread across Gmail, Chat, Meet, Messages, Voice, Contacts and social-media systems.

15.1 One conversation graph, many providers

A conversation is a portable event graph rather than a folder trapped in one account. It can contain:

- asynchronous messages;
- real-time chat;
- voice and video sessions;
- shared files;
- calendar proposals;
- decisions and action items;
- credentials and signatures;
- moderation events;
- retention or deletion policies.

Providers compete on interface, reliability, encryption, storage and support. The conversation identity and export format remain portable.

15.2 Community communication

Communities need more than private messages. TOM Link supports:

- public and private rooms;
- neighbourhood or institutional channels;
- issue-based forums;
- deliberation and voting links;
- event coordination;
- emergency notices;
- creator/follower publication;
- moderation constitutions;
- archival and ephemeral modes.

Matrix and ActivityPub provide existing technical foundations for federation, but community governance, anti-abuse operations and usability remain major work. [W3] [W5]

15.3 Safety without one global censor

Moderation has layers:

personal controls

room/community rules

provider rules

jurisdictional obligations

network-level abuse signals

independent appeals

Illegal or dangerous material cannot be treated as a mere preference. At the same time, no single commercial recommendation system should invisibly determine every community's public conversation. Rules and enforcement events must be attributable and appealable where possible.

16. TOM Workspace - documents, files, projects and time

TOM Workspace replaces the functional surface of Drive, Docs, Sheets, Slides, Forms, Sites, Keep, Tasks and Calendar.

16.1 File and object model

A document is not merely a binary blob. It has:

- content identity;
- editor and contribution lineage;
- permissions;
- comments and decisions;
- referenced evidence;
- publication status;
- templates and schemas;
- derived formats;
- retention and deletion policy;
- links to projects, people, services and manufactured artifacts.

Collaborative editing can use existing open document formats and conflict-free or versioned change systems. TOM supplies the authoritative snapshots, permissions, decisions and releases.

16.2 Work graphs

Projects are graphs of goals, tasks, dependencies, resources, evidence and decisions. A task can be completed by a person, AI agent, software process, external service, robot, laboratory or public institution. Each completion has a typed outcome and verifier.

16.3 Calendar as negotiated time

A calendar event can express:

- participants and roles;
- time windows rather than only fixed times;
- accessibility and location constraints;
- travel time;
- resource availability;
- privacy;
- agenda, decisions and follow-up;
- recurrence and expiry;
- external booking conditions.

Scheduling agents propose; participants and formal rules accept.

17. TOM Atlas - maps, earth, mobility and travel

TOM Atlas replaces Maps, Earth, Street View, Waze, Flights and Travel as a federation of geographic data, routing and service operators.

17.1 The map is a commons of claims

OpenStreetMap already shows how local contributors can maintain open data about roads, trails, cafes, railway stations and many other features. [W6] TOM Atlas adds typed provenance and service layers:

geographic feature
 source and observation date
 licence
 confidence or dispute
 accessibility
 operating hours
 hazards and restrictions
 public-service links
 mobility modes
 repair and infrastructure history

17.2 Selectable routing goals

A route is not universally “best”. Profiles may prioritise:

- fastest;
- lowest cost;
- wheelchair access;
- safest cycling infrastructure;
- fewest transfers;
- lowest emissions;
- quietest or least polluted;
- child-friendly;
- emergency access;
- scenic or cultural value;
- robustness under disruption.

The system displays trade-offs instead of hiding them in a global default.

17.3 Place and community knowledge

Local communities can publish verified service directories, public facilities, mutual-aid resources, cultural histories, repair shops, accessibility notes and emergency plans. A place page becomes a living evidence graph rather than a commercial review wall.

17.4 Travel and booking

Travel search separates:

- schedule data;

- price and availability;
- operator reliability;
- accessibility;
- baggage and cancellation rules;
- emissions and alternatives;
- sponsored offers;
- booking service fees.

A TOM Agent may assemble an itinerary, but purchase requires explicit authority and the receipt records which options were considered.

18. TOM Media - video, music, books, news, photos and culture

TOM Media replaces the functional surface of YouTube, YouTube Music, YouTube TV, Play Books, Photos, News, Arts & Culture and creator-distribution systems.

18.1 Creator sovereignty

A creator can publish through any compatible host while retaining:

- stable identity;
- catalogue metadata;
- subscriber/follower relationships;
- rights and licence records;
- revenue and sponsorship disclosures;
- moderation status;
- archival copies;
- migration path.

Discovery profiles can prioritise subscriptions, local culture, educational value, novelty, independent creators or other visible objectives.

18.2 Recommendation as user-selected curation

Recommendation remains useful but becomes plural:

- personal local recommender;
- creator-curated channel;
- community editorial board;
- public broadcaster profile;
- educational curriculum;
- commercial entertainment profile;
- chronological feed;
- intentionally diverse discovery profile.

A recommendation receipt identifies the profile and major factors. “Because the platform predicts engagement” is no longer the only invisible default.

18.3 News as claim and event graph

A news item should distinguish:

- reported event;
- primary documents;
- observations and witnesses;
- outlet framing;
- corrections;
- related historical events;
- disputed claims;
- funding or ownership disclosures;
- synthetic media provenance.

AI summaries can compare reporting but must link every consequential assertion to the source set.

18.4 Personal photos and memory

Photos and videos remain in the person's chosen vault. AI may organise them locally or under explicit permission. Shared albums are portable collaboration objects rather than permanent surrender to one platform.

19. TOM Learn and TOM Scholar - education, research and translation

TOM Learn combines the functions of Classroom, Scholar, Notebook-style research assistance, Translate and public educational resources.

19.1 Personal learning paths

A learning request becomes:

current verified knowledge
+ desired capability
+ available time and language
+ accessibility needs
+ budget
+ preferred learning modes
-> candidate curricula
-> prerequisite graph
-> evidence of completion
-> portable credentials

The system should not optimise only for course completion or commercial conversion. It should show which skills were demonstrated and which remain uncertain.

19.2 Open research graph

OpenAlex and Wikidata demonstrate the feasibility of open structured research and knowledge graphs. [W7] [W9] TOM Scholar adds:

- exact paper, dataset, software and method identities;
- citation and derivation paths;
- replication attempts;
- retractions and corrections;
- result contradictions;
- benchmark versions;
- machine-readable claims;
- links from research to manufacturing or public policy.

19.3 Research AI

A research agent can:

- build literature maps;
- extract candidate claims;
- compare methods;
- identify missing evidence;
- draft reproducible workflows;
- call external simulators;
- generate hypotheses;
- prepare review tables.

It cannot promote its synthesis as scientific truth without independent evidence.

19.4 Translation and cultural context

Translation services should preserve:

- source text;
- target text;
- model or human translator identity;
- uncertainty or alternative translations;
- terminology glossary;
- cultural or legal context;
- corrections.

Public institutions and communities can maintain terminology packs for underserved languages without waiting for one company's commercial priority.

20. TOM Market, TOM Wallet and TOM Work

TOM Market replaces Shopping, Pay, Wallet, Finance, Business Profile, Merchant systems and parts of advertising-driven commercial discovery.

20.1 Product and service discovery

A product result can include:

- seller identity and jurisdiction;
- total delivered cost;
- warranty and return rules;
- material and manufacturing provenance;
- repairability and spare parts;
- energy and environmental profile;
- compatibility with owned devices or systems;
- safety certifications;
- independent test evidence;
- known recalls;
- sponsored status.

Price remains important, but it is not the only ranking factor.

20.2 Wallet and payments

TOM Wallet stores or references:

- payment credentials;
- identity and eligibility credentials;
- tickets and passes;
- licences;
- receipts;
- warranties;
- membership and access rights;
- public benefits;
- product passports.

Actual payment networks remain external regulated services. TOM records the permission, terms, transaction identity and receipt.

20.3 Work and livelihood

A job and work network should support:

- verified skills and work samples;
- portable references;
- transparent eligibility criteria;
- salary and condition disclosure;
- accessibility and care constraints;
- union and professional resources;
- project and cooperative work;
- apprenticeship and learning paths;

- anti-discrimination audit;
- appeals for automated screening.

AI may match and explain; employers or accountable rules decide. Hidden personality scoring should not become the default gate to livelihood.

20.4 Small business without platform captivity

A small organisation can publish a portable business profile containing location, services, credentials, opening times, accessibility, prices, reviews, product passports and contact routes. It can move hosting or payment provider without losing discoverability or accumulated verified history.

21. TOM Civic - public services, law and democratic participation

TOM Civic is the part of the network most unlike a commercial search engine. It connects discovery to accountable public action.

21.1 Public-service navigator

A person can ask:

- Which benefits or services might apply?
- What evidence is required?
- Which deadline and authority govern the application?
- Why was a decision made?
- How can I correct a record or appeal?
- Which local office or human advocate can help?

Rules are versioned and jurisdiction-specific. An AI may explain and help prepare a submission, but the accountable public authority remains visible.

21.2 Law and policy graph

A law or policy page links:

source text
jurisdiction and effective dates
amendments
implementing regulations
court or review decisions
budget and responsible institution
plain-language explanations
impact evidence
public consultations
appeal and complaint routes

The system should show uncertainty when a rule is disputed or interpretation depends on professional advice.

21.3 Democratic participation

TOM Civic can support:

- verified public notices;
- consultation submissions;
- participatory budgeting;
- petition and initiative workflows;
- meeting agendas and minutes;
- representative contact;
- conflict-of-interest and funding disclosures;
- proposal comparison;
- public evidence repositories;
- post-decision outcome tracking.

It does not replace political judgement with deterministic machinery. It makes the inputs, rules, commitments and outcomes harder to hide.

21.4 Digital public infrastructure

UNDP describes digital public infrastructure work in terms of digital ID governance, standards, scalable digital solutions and whole-of-society partnerships. [H2] TOM Commons can function as an application and evidence layer over existing public identity, payment and data-exchange rails, provided rights, inclusion and redress are built in.

22. TOM Health, TOM Home, TOM Family and TOM Crisis

22.1 Health navigation, not autonomous medicine

TOM Health can help a person:

- find qualified care;
- understand and organise records;
- compare treatment information and guidelines;
- prepare questions;
- manage appointments and referrals;
- share selected records;
- track consent;
- discover trials or support resources;
- audit how an AI explanation was produced.

It must not present the current TOM kernel as a doctor, diagnosis engine or treatment authority. Clinical decisions remain with qualified professionals and regulated systems.

22.2 Personal and family coordination

TOM Family supports:

- shared calendars and care tasks;
- school and childcare records;
- delegated permissions;
- emergency contacts;
- household budgets;
- family archives;
- age-appropriate controls;
- elder care coordination;
- guardianship and consent records.

Children's data receives special protection. Family coordination must not become permanent surveillance.

22.3 Home and devices

TOM Home replaces parts of Home, Nest, Cast, Find Hub and device ecosystems. It represents devices as portable, inspectable capabilities:

device identity
owner and household permissions
sensors and actuators
firmware and update lineage
local automation rules
external cloud dependencies
energy use
repair and end-of-life path

Automations run locally where possible. Safety-critical devices retain independent physical interlocks.

22.4 Crisis mode

Crisis services need a special mode:

- authoritative alerts separated from ordinary media;
- low-bandwidth and offline operation;
- multilingual and accessible presentation;
- shelter, food, medical and transport availability;
- family check-in;
- community resource reporting;
- rumours and unverified reports clearly labelled;
- privacy-preserving location sharing;
- post-event audit and correction.

No commercial ranking or sponsorship should alter emergency results.

23. TOM Compute and TOM Builder - cloud and developer infrastructure

TOM Compute replaces the functional surface of Cloud, Firebase, developer tools, analytics and parts of the application ecosystem.

23.1 Portable workloads

A service package declares:

- source and build identity;
- runtime requirements;
- data access permissions;
- resource limits;
- network interfaces;
- geographic or jurisdiction constraints;
- logs and observability;
- backup and recovery;
- cost model;
- security and conformance evidence.

Operators compete to run the same package. Public services can require reproducible builds and open exit paths.

23.2 Developer commons

TOM Builder provides:

- open SDKs for needs, evidence, identity, ranking, service and action records;
- local test harnesses;
- reference clients and servers;
- schema registries;
- conformance suites;
- reproducible packaging;
- public sample datasets;
- adapter templates;
- accessibility testing;
- security review profiles.

23.3 Analytics without behavioural enclosure

Organisations still need metrics. TOM Analytics separates:

- aggregate service quality;
- user-controlled personal analytics;
- public outcome measures;
- security and reliability telemetry;

- commercial conversion analytics.

Cross-service tracking requires explicit consent and purpose. Public-interest services should be measured by whether needs are met, not only by session length or clicks.

24. TOM Foundry, Biofoundry, Habitat and Frontier

This is where the companion links directly to the Manufacturing Horizons report.

24.1 TOM Foundry

TOM Foundry turns a need into a proof-carrying product chain:

need

- > requirements
- > candidate designs
- > simulation and independent checks
- > material and supplier compatibility
- > approved process plan
- > machine execution
- > measurement and inspection
- > release or rejection
- > maintenance, repair and recycling lineage

Search results can therefore end in a repair instruction, spare part, local service bureau, certified software image, electronic assembly or manufactured object.

24.2 TOM Biofoundry

TOM Biofoundry applies the same evidence architecture to regulated biological research:

research question

- > candidate biological definition
- > biosafety and ethics gate
- > non-authoritative simulation
- > approved experimental protocol
- > instrument and batch observations
- > independent analysis
- > accept, reject or ambiguous result
- > versioned lineage

It does not make biology deterministic and does not bypass qualified oversight. The domain remains measurement-driven and highly regulated.

24.3 TOM Organ

The Manufacturing Horizons report describes a long path from digital design to nonimplantable tissue, perfused modules, bioartificial support, implantable auxiliary organs and eventually whole or novel organs. TOM Commons adds the social and service infrastructure:

- patient and donor rights;
- consent and privacy;
- research discovery;
- trial and eligibility information;
- protocol and evidence registries;
- manufacturing lineage;
- clinical and regulatory review;
- long-term outcomes;
- equitable access and allocation;
- appeals and public governance.

The existence of a deterministic record does not establish that an organ is safe or effective.

24.4 TOM Habitat and Frontier

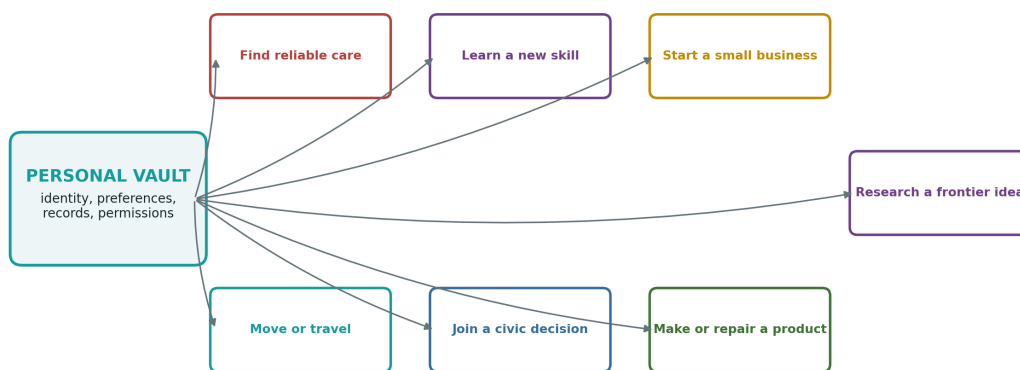
At longer horizons, the same network coordinates:

- resilient housing and infrastructure;
- circular manufacturing;
- autonomous laboratories;
- living materials;
- orbital microfactories;
- closed ecological systems;
- planetary observation;
- long-duration settlement;
- intergenerational knowledge archives.

These are frontier programmes, not current TOM capabilities. Their value in this report is architectural continuity: the identity and evidence system used for today's repair part can scale conceptually to a future habitat component or engineered tissue without altering the kernel.

Part IV - Human journeys across the network

ONE LIFE, ONE PORTABLE CONTEXT, MANY INDEPENDENT SERVICES



The person can change providers without abandoning identity, history or rights - the relationship graph belongs to the person and the commons.

Figure 8: A portable personal context can support many independent life services without one provider owning the person.

The following journeys show how the modules cooperate. They are not promises of current deployment. They are design tests: if the architecture cannot handle ordinary life events clearly and fairly, it is not a credible replacement for a platform ecosystem.

25. Journey 1 - find reliable care without surrendering a life profile

A person reports a persistent symptom and asks where to seek appropriate care. TOM Health first clarifies that it can provide navigation and information, not diagnosis. The person authorises access to location, language, insurance or public coverage, accessibility requirements and preferred appointment windows. Medical history remains private unless the person chooses to share it with a selected provider.

The query searches public provider registries, qualification credentials, service availability, travel time, accessibility, accepted coverage and verified complaint or quality data. It distinguishes official provider information, independent quality evidence, patient experience and sponsored place-

ment. An AI agent can explain options and prepare questions, but it cannot silently exclude a provider or recommend treatment as authoritative.

The result bundle might include:

three eligible clinics
 travel and accessibility comparison
 credential verification
 appointment availability
 expected cost and coverage uncertainty
 source dates
 plain-language preparation notes
 record-sharing permissions
 human review / emergency guidance

If the person books, the appointment event records only the necessary information. The clinic receives a selective presentation of credentials and records. After care, the person can import a visit summary into the vault, correct errors and decide whether anonymised outcomes may support public quality research.

Replacement value: search, maps, reviews, calendar, mail, wallet, records and AI explanation cooperate without one advertising profile controlling the journey.

26. Journey 2 - turn curiosity into a verified learning path

A person wants to become competent in electronics design while working part time. TOM Learn asks for the desired outcome, existing knowledge, available hours, language, accessibility, budget and preferred modes. Verified prior learning can be shared selectively; informal experience can be represented as a claim requiring assessment.

The system constructs a prerequisite graph across open courses, local colleges, mentors, textbooks, simulation tools and project challenges. Ranking can prioritise low cost, recognised credentials, hands-on work, local support or fastest completion. The AI tutor explains concepts and drafts exercises, while deterministic assessments or human review produce evidence of competence.

A learning path is not a single vendor subscription. It may combine:

- an open mathematics course;
- a community workshop;
- circuit simulation;
- a public library textbook;
- a mentor session;
- a small proof-carrying hardware project;
- a portable credential.

When a course disappears or provider changes terms, the path can substitute a compatible resource without erasing completed evidence.

Replacement value: classroom, scholar, notebook, video, books, search, calendar and credentials become a portable competence graph.

27. Journey 3 - find dignified work and keep the evidence

A worker wants a role compatible with verified skills, a minimum salary, caregiving hours and wheelchair access. The query shares only those constraints. Employers publish machine-readable role definitions: responsibilities, pay range, schedule, location, accessibility, required credentials, decision process and appeal contact.

Matching algorithms and AI agents may propose opportunities, but the ranking profile is visible. The worker can prefer cooperative employers, training opportunities, commute limits, remote work or sector mission. Automated screening decisions must disclose the criteria used and permit correction of erroneous records.

The worker's portfolio contains content-addressed work samples and credentials. References are portable, revocable statements rather than social capital trapped inside one platform. Applications are separate permissioned events; the agent cannot mass-submit without authorisation.

If a job is rejected, the system can preserve the employer's stated reason, legal obligations and appeal path. Aggregate public metrics can reveal patterns without exposing individual applications.

Replacement value: search, business profiles, communication, documents, calendar, identity and AI matching become a transparent livelihood service rather than an opaque engagement funnel.

28. Journey 4 - start a small business without becoming platform-dependent

A person wants to offer repair services. TOM Market helps create a portable business profile with location, service area, skills, credentials, pricing model, accessibility, appointment method and warranty. The profile can be hosted by a cooperative, municipality, trade association or commercial provider while retaining stable identity.

TOM Atlas makes the service locally discoverable. TOM Link handles enquiries. TOM Calendar negotiates appointments. TOM Wallet records quotes, payments and receipts. TOM Workspace stores repair reports. TOM Foundry links replaced parts to product passports. Reviews become signed experience claims with anti-retaliation and dispute mechanisms, not anonymous stars alone.

Sponsored discovery is allowed but separated from organic local relevance and evidence. The business can change web host, payment provider or booking tool without losing its verified history.

A future AI agent may draft quotes, diagnose from customer-provided evidence or source parts, but the business remains accountable and approves commitments.

Replacement value: business profile, maps, search, ads, mail, calendar, payments, analytics and manufacturing lineage become an open small-business stack.

29. Journey 5 - move house through one rights-preserving service graph

Moving combines housing search, maps, work, schools, healthcare, utilities, transport, finance, documents and public registrations. Today these are usually separate portals with repeated forms and inconsistent records.

In TOM Commons, the person declares housing requirements and a disclosure ceiling. The query combines verified listings, total cost, accessibility, transit, hazard and energy data, landlord or seller credentials, local services and community information. Commercial listings, public housing and cooperative options can appear in one eligible set without hidden sponsorship.

An agent prepares a move plan:

- housing applications
- viewings and travel
- budget scenarios
- utility transfer
- school or care coordination
- address changes
- moving services
- product inventory
- insurance questions

Each action remains separately authorised. The person's identity and documents are presented selectively. A provider cannot demand unrelated browsing history merely because the user searched for a home.

Replacement value: maps, travel, search, wallet, workspace, calendar, civic services and AI planning converge around a real human outcome.

30. Journey 6 - navigate benefits and public services with reasons and appeal

A person loses income and asks what support may apply. TOM Civic retrieves the relevant jurisdiction, effective rules, service availability and official source text. An AI proposes a plain-language interpretation and asks only for facts required by the selected rules.

The system produces:

- possible programmes;
- why each might apply;
- uncertain or missing facts;
- required documents;
- application deadlines;
- expected process;
- human assistance contacts;
- privacy and record-use terms;
- appeal rights.

The actual public authority remains responsible for the decision. If a deterministic eligibility rule is used, the result receipt shows the rule version and input fields. If discretion or professional judgement is involved, that fact is explicit.

A rejected application generates a structured reason and appeal object. The person can correct an erroneous address, income record or family status without rebuilding the entire application.

Replacement value: search and AI become accountable public navigation rather than a maze of unofficial pages and advertisements.

31. Journey 7 - understand a public decision and participate

A community learns that a major infrastructure project is proposed. TOM Civic creates a public evidence graph linking the legal notice, maps, budgets, environmental reports, design alternatives, meeting dates, responsible agencies, contractor disclosures and public comments.

Different ranking profiles allow residents to see the newest official documents, strongest independent analyses, local impacts or unanswered questions. AI summaries are generated from the same source graph and can be compared across models. Conflicts are preserved rather than averaged away.

Residents submit comments through signed or anonymous channels as permitted. A deliberation space records proposals, amendments, support and objections. The final public decision links back to the evidence and commitments it relied on. Later outcome measurements show whether promised benefits and mitigations occurred.

Replacement value: search, maps, news, video, documents, meetings and civic action become one accountable public process.

32. Journey 8 - respond to a disaster with local knowledge and public authority

During a flood or wildfire, TOM Crisis switches to a constrained profile. Official alerts, verified shelters, road closures, medical help, water and food distribution, transport and family check-ins are prioritised. Sponsored content is disabled. The client supports low bandwidth, offline caching and multilingual accessibility.

Community members can submit observations, but unverified reports are marked and cross-checked. Open map data and local knowledge can update routes faster than a distant central team, while official restrictions remain identifiable. A person may share approximate location temporarily with selected family or emergency services and revoke it after the event.

After the crisis, the event graph supports accountability: which alerts were issued, which resources were available, which claims were false and where infrastructure failed.

Replacement value: maps, search, messaging, public alerts and community coordination operate

as a public emergency utility.

33. Journey 9 - conduct reproducible research from question to result

A researcher asks whether a material or algorithm performs better under a declared condition. TOM Scholar retrieves papers, datasets, code, retractions, replication attempts and competing methods. The researcher creates a hypothesis record and a reproducible workflow.

External tools perform simulation, statistical analysis or model training. TOM records exact versions, input hashes, parameters, environment and output identities. Independent oracles or alternative implementations attempt to falsify the result. An AI assistant helps search, code and write, but its contributions remain attributable.

The publication bundle contains:

- question and hypothesis
- source literature graph
- dataset and licence
- analysis definitions
- software and environment
- results
- counterexamples
- limitations
- peer review and revisions

If a later study contradicts the result, it creates a linked record rather than silently overwriting history.

Replacement value: scholar, notebooks, cloud, storage, AI and publication become a proof-carrying research environment.

34. Journey 10 - manufacture and repair a physical product

A household needs a replacement bracket for an appliance whose manufacturer no longer supplies the part. TOM Query identifies the product passport, compatible part definition, material requirements, safety constraints and authorised repair paths. A local fabricator publishes machine capability and quality credentials.

TOM Foundry creates a candidate manufacturing package:

- part design and version
- compatibility with appliance model
- material specification
- process plan
- machine profile
- inspection dimensions
- liability and warranty terms
- price and delivery

Simulation and human review verify fit. The person approves the order. The fabricator produces the part and attaches measurement evidence. The repaired appliance becomes a descendant product record with the new part and date. Future owners can see what changed.

Replacement value: search, shopping, maps, payment, documents and manufacturing become a circular repair network instead of forced replacement.

35. Journey 11 - move a biological idea through a safe research pathway

A research team proposes a tissue model or engineered living material. TOM Scholar finds prior work, protocols, safety guidance and known failures. The proposal is classified as biological research and cannot move directly to fabrication.

The pathway requires:

- institutional authority;
- biosafety and ethics review;
- defined organism, material or tissue scope;
- containment and disposal plan;
- approved protocol;
- instrument and batch identities;
- measurement plan;
- independent analysis;
- explicit stop and escalation conditions;
- publication and access controls.

External laboratory systems produce observations. TOM Biofoundry records them and supports comparison against the accepted protocol. The result may be accepted as a research finding, rejected or remain ambiguous. Clinical or environmental use would require a separate, much stronger promotion chain.

Replacement value: search, research, workflow, laboratory automation and manufacturing lineage are connected without treating a web answer as permission to create life.

36. Journey 12 - plan a long-term habitat or space manufacturing programme

A consortium explores a resilient settlement, orbital factory or closed habitat. TOM Commons becomes the information and coordination layer:

- public goals and constraints;
- rights of participants;
- scientific models;
- materials and manufacturing definitions;
- energy, water, food and waste systems;
- health and emergency plans;
- governance and appeals;
- supply-chain and mission lineage;
- independent audits;
- public reporting.

Multiple simulations and engineering teams contribute proposals. The network preserves disagreement and prevents one impressive model from becoming unquestioned authority. The Manufacturing Horizons framework provides the proof-carrying fabrication chain; TOM Commons provides collective knowledge, consent and decision infrastructure.

Replacement value: the same human-needs interface that finds a bus route can, at a radically larger scale, coordinate a multi-decade frontier programme without changing its constitutional principles.

Part V - From information retrieval to civilisation-scale manufacturing

37. Discovery becomes a production primitive

The historic web-search model ends at information or a commercial click. TOM Commons extends the terminal set:

page
 answer
 conversation
 document
 course
 service appointment
 public application

purchase
 software build
 manufacturing order
 scientific experiment
 biological research protocol
 community decision

The system can therefore connect a need to an outcome while preserving the difference between finding, recommending, deciding and acting.

37.1 The product passport as a search result

A search for a device or part can return not only a shop page but its product lineage:

- design version;
- manufacturer and facility;
- material and component sources;
- certifications;
- software and firmware;
- repair manuals;
- compatible replacements;
- known failures and recalls;
- energy and environmental information;
- end-of-life routes.

This makes search a circular-economy tool.

37.2 Manufacturing capability as a public index

Factories, workshops, service bureaus and laboratories can publish capability records:

processes
 materials
 size and tolerance envelopes
 quality systems
 machine identities
 availability
 location
 price model
 certifications
 accepted design formats
 inspection capability
 restricted or prohibited work

A design agent can find compatible operators without sending sensitive design data to every provider.

38. Physical and biochemical horizons in the commons

The earlier Manufacturing Horizons report includes conventional fabrication, advanced materials, molecular systems, engineered living materials, tissues, organs, synthetic organisms, habitats and off-world industry. TOM Commons supplies five missing social layers.

38.1 Discovery

Researchers, manufacturers, regulators, patients, communities and funders can find definitions, facilities, evidence and unresolved problems.

38.2 Access

Eligibility, cost, location, capacity and rights are represented explicitly. Frontier technology should not be governed only by those who own a closed platform or laboratory.

38.3 Governance

Biological and planetary interventions require consent, ethics, biosafety, ecological review and public accountability. These are first-class service and decision records, not afterthought PDFs.

38.4 Longitudinal evidence

A tissue, organ, living material or habitat component may evolve over years. The system links design, manufacture, implantation or deployment, monitoring, repair and outcome.

38.5 Equitable allocation

Scarce treatments, manufacturing capacity or emergency resources require visible allocation principles, appeals and public outcome audit. A private ranking algorithm must not decide quietly.

39. Frontier modules

39.1 TOM Materials

A global evidence graph for materials links composition, structure, process, simulation, measured properties, uncertainty, suppliers, standards, environmental effects and product use. AI and autonomous laboratories propose candidates; definitions are promoted only after evidence.

39.2 TOM Organ

A future organ network links patients, researchers, tissue sources, consent, cell lines, scaffolds, protocols, bioreactors, measurements, regulatory review, transplantation, monitoring and long-term outcomes. The system can support fair discovery and evidence without claiming that current organ manufacturing is solved.

39.3 TOM Life

Synthetic organisms or engineered ecological functions are treated as restricted, high-consequence definitions. The network records containment, intended function, failure modes, ecological interactions, monitoring and revocation or remediation plans. No open search result automatically becomes an executable biological recipe.

39.4 TOM Habitat

Buildings and infrastructure become searchable, repairable, adaptive product systems with material passports, digital twins, energy/water models, accessibility, maintenance and community governance.

39.5 TOM Space

Space manufacturing and settlement programmes use the same evidence and governance architecture across mission design, remote operations, component lineage, communication delay, local autonomy and intergenerational knowledge.

Future horizon

A far-future TOM Commons could act as a civilisation memory and coordination layer linking Earth communities, orbital industry and distant settlements. That is compatible with ordinary physics and communication delay, but it remains a long-horizon institutional concept. The current kernel does not provide global consensus, autonomous scientific truth or self-sufficient planetary infrastructure.

Part VI - A buildable transition

40. Start without money: the laptop prototype

A useful first version does not require a new data centre, phone, robot, factory or laboratory. It requires a coherent local application and carefully selected open sources.

40.1 Minimum viable TOM Commons

local desktop or web client
 personal encrypted vault
 canonical need records
 open web / research / map adapters
 visible ranking profiles
 one or more optional language models
 answer receipts and source graphs
 portable identity and passkey login
 Matrix or email bridge for communication
 basic service registry
 TOM execution and lineage layer

40.2 Initial datasets and services

Need	Present open foundation	First prototype
Web discovery	Common Crawl and public websites	Index a bounded topic or language subset
Structured knowledge	Wikidata	Evidence-linked entity and claim explorer
Maps	OpenStreetMap	Local services, accessibility and repair map
Research	OpenAlex and open repositories	Literature map with reproducibility links
Communication	Matrix and ordinary email bridges	Portable project rooms and notifications
Personal data	Solid-like Pod or local encrypted store	User-controlled documents and context
Credentials	Verifiable Credentials, DIDs, WebAuthn	Portable membership and qualification proof
AI	Local or API models	Proposal layer with retrieval and claim receipts
Manufacturing	Public CAD/process examples and local makers	Proof-carrying repair-part workflow
Civic	One municipality's open data	Public-service and decision navigator

The prototype should be intentionally narrow: one city or region, one research domain and one manufacturing demonstration. The success criterion is not search volume. It is whether people can understand, transfer and challenge the result.

41. Reference architecture

41.1 Client

The client provides:

- need capture;
- vault permissions;
- query profile selection;
- result and evidence display;
- communication and document views;
- action confirmations;
- appeals and corrections;
- offline mode;
- accessibility and translation.

41.2 Index gateway

The gateway queries multiple indexes, normalises result records and preserves operator identity. It cannot silently merge away disagreement.

41.3 Evidence engine

The evidence engine resolves claims, provenance, contradictions, versions and domain-specific checks. It produces a graph and a signed receipt.

41.4 AI orchestrator

The orchestrator sends only permitted context to selected models, captures prompts and outputs, extracts claims and returns proposals to the evidence engine.

41.5 Service/action broker

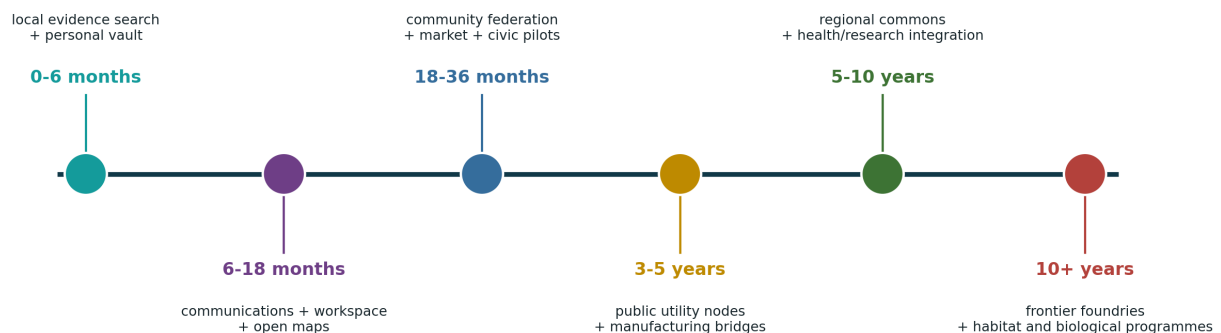
The broker handles messages, bookings, payments, applications, tool calls and manufacturing requests. Every consequential operation requires a policy and permission check.

41.6 TOM authority node

The TOM node stores the canonical definitions and executes deterministic decisions and lineage. Under the current repair boundary, authoritative writes are coordinated on one host. Replication and federation require an explicit external protocol.

42. Roadmap

TRANSITION ROADMAP: REPLACE FUNCTIONS, NOT PEOPLE



Every phase must preserve portability, user data exit, explicit ranking, independent audit and the frozen TOM kernel boundary.

Figure 9: A staged transition replaces functions one by one while preserving data portability and the frozen kernel.

Phase 0 - constitutional prototype, 0 to 6 months

Deliver:

- Human Needs Genome schema;
- local personal vault;
- bounded open-source index;
- two visible ranking profiles;
- cited AI answer mode and no-AI mode;
- query receipts;
- import/export;
- one local-service catalogue;
- one proof-carrying manufacturing demonstration;
- accessibility review;
- public threat model.

Exit test: a user can move the vault and ranking profile to a second operator and reproduce a selected result receipt.

Phase 1 - communication and workspace, 6 to 18 months

Deliver:

- federated communication;
- collaborative documents and calendar;
- portable contacts;
- credentials and passkeys;
- community moderation constitution;
- public research and map connectors;
- service-booking pilot;
- independent security audit.

Exit test: a community can change hosting providers without losing identities, conversations, documents or service definitions.

Phase 2 - city or regional commons, 18 to 36 months

Deliver:

- public-service navigator;
- local business and repair network;
- community map and event layers;
- public meeting and consultation graph;
- member governance;
- cooperative payment/receipt integration;
- several independent index and AI providers;
- formal appeals process.

Exit test: the region can operate core services during the failure or withdrawal of one provider.

Phase 3 - public utility and manufacturing bridge, 3 to 5 years

Deliver:

- large public-interest web and research indexes;
- interoperable vault providers;
- procurement and product passport standards;
- digital-twin and manufacturing capability registry;
- reproducible software and device release gates;
- specialist health and legal information services;
- cross-operator conformance and outcome benchmarks.

Exit test: discovery can produce a verified software or physical product through independent providers with end-to-end lineage.

Phase 4 - multinational federation, 5 to 10 years

Deliver:

- multilingual public indexes;

- jurisdiction-aware credentials and services;
- cross-border mobility, education and research records;
- distributed governance and consensus layer external to the fixed kernel;
- public-interest AI models;
- crisis interoperability;
- large-scale funding and anti-capture institutions.

Exit test: a person can migrate across regions and providers while retaining rights, credentials and history.

Phase 5 - frontier programmes, 10 years and beyond

Possible programmes:

- autonomous laboratories under explicit authority;
- regional circular manufacturing;
- advanced tissue and organ evidence networks;
- living materials and habitat research;
- orbital manufacturing coordination;
- long-duration settlement knowledge systems.

Exit tests are domain-specific and require independent physical, clinical, ecological and regulatory evidence.

43. Metrics that reflect human outcomes

A replacement platform should not judge itself primarily by queries, clicks, time spent or ad revenue.

Domain	Better outcome measures
Search	source diversity, evidence traceability, correction rate, task success, portability
Communication	delivery reliability, provider migration, abuse response, user control
Education	demonstrated competence, accessibility, completion without debt, credential portability
Health navigation	time to appropriate care, record accuracy, consent, avoidable duplication
Public services	successful access, understandable reasons, appeal resolution, inclusion
Work	transparent conditions, skill recognition, fair screening, livelihood stability
Market	total cost, repairability, dispute resolution, small-provider discoverability
Maps/mobility	accessibility, local data quality, route resilience, community correction
Media	creator portability, user-chosen recommendation, provenance, cultural diversity
Manufacturing	defect rate, reproducibility, repair, material lineage, circularity
Governance	member participation, audit findings, provider concentration, successful exits
AI	citation coverage, detected contradictions, human override, reproducible tool use

44. Risk register

A post-platform system can fail in new or familiar ways. The architecture must assume adversaries, accidents and institutional capture.

Risk	Failure mode	Required response
Re-centralisation	One index, vault or AI provider gains dominant network power	Portability tests, concentration limits, public alternatives and forkability

Risk	Failure mode	Required response
Governance capture	Wealthy members, governments or insiders control rules	Transparent finance, term limits, multi-stakeholder chambers and independent audit
Data poisoning	Malicious sources contaminate open indexes	Provenance, anomaly detection, independent indexes and source reputation as explicit evidence
False determinism	A wrong formal rule appears trustworthy because it is reproducible	Independent oracle, appeals, regression evidence and separation of execution truth from world truth
Privacy correlation	Credentials and queries are linked across contexts	Selective disclosure, pseudonyms, data minimisation and local processing
Identity exclusion	People without documents or modern devices lose access	assisted channels, offline credentials, community attestation and public service obligations
Ranking manipulation	Operators bias results or sponsors influence organic ranking	published profiles, signed receipts, independent replay and sponsorship separation
Moderation abuse	Rules suppress legitimate speech or fail to stop abuse	layered governance, transparency, appeals and community choice within rights constraints
AI hallucination	Fluent synthesis invents or distorts claims	source-linked claim extraction, model comparison, no-AI mode and human escalation
Distributed inconsistency	Federated nodes disagree on authoritative state	explicit consensus layer; do not pretend same-host TOM locking solves it
Cost escalation	Open infrastructure becomes unaffordable	public funding, efficient local indexes, caching and plural operators
Accessibility failure	Complex evidence interfaces exclude users	progressive disclosure, co-design and conformance testing
Legal fragmentation	Jurisdictions impose conflicting duties	jurisdiction-tagged rules, local operators and visible conflicts
Biological misuse	Research discovery lowers barriers to harmful action	restricted access, non-operational public descriptions, biosafety review and monitoring
Physical harm	Agent or manufacturing action bypasses safety systems	external certified controls, human approval and staged deployment
Permanent records	Lineage conflicts with privacy or rehabilitation	scoped retention, revocation, sealed records and lawful deletion mechanisms
Commons neglect	Public data becomes stale or under-maintained	funded stewardship, quality metrics and local contribution pathways

45. What should remain human

TOM Commons is not a plan to automate away society. Some functions should remain irreducibly human or institutionally accountable:

- choosing collective values;
- care, empathy and relationship;
- artistic and cultural meaning;
- professional judgement under uncertainty;
- moral responsibility;
- political negotiation;
- legal adjudication and due process;
- consent;
- high-consequence scientific and medical oversight;
- deciding what future is desirable.

The network can make evidence, alternatives and commitments visible. It cannot compute the one objectively correct civilisation.

46. What should be replaced

The strongest candidates for replacement are mechanisms, not people:

- repeated form filling for facts already held under consent;

- account captivity;
- opaque search and recommendation objectives;
- hidden sponsored ranking;
- AI answers detached from source receipts;
- closed social graphs;
- inaccessible public portals;
- irreversible provider lock-in;
- software releases without reproducible evidence;
- product histories that disappear after sale;
- manufacturing decisions hidden in spreadsheets;
- research claims detached from data and code;
- unsupported automated decisions without appeal;
- bureaucratic verification duplicated across institutions.

47. Final formal definition

TOM principle

TOM Commons is a proposed federated human-needs discovery, communication, service, evidence and manufacturing network built above the frozen TOM seeded substrate. A person or community expresses a typed need and controlled context. Independent indexes, institutions, service providers, models, simulators and machines contribute evidence or proposals. Support and compatibility determine eligibility; a visible ranking profile orders candidates; AI may explain or plan without becoming hidden authority; consequential actions require explicit permission; decisions, services, fabrication and outcomes create content-addressed lineage. Identity, data, social graphs, credentials and histories remain portable across providers. Commercial sponsorship is visibly separated from organic evidence. Public, cooperative and competitive operators share open standards, rights obligations, conformance tests and independent audit. The network may extend from ordinary search and communication to public services, research, proof-carrying manufacturing, biological programmes and planetary infrastructure without changing the canonical TOM kernel.

The civilisational ambition is not a universal corporate account. It is a universal **right to ask, understand, act, move, create and appeal without surrendering one's identity and future to one opaque intermediary**.

Appendix A - Functional mapping from the present Google surface to TOM Commons

Google's official products page is the source for the product names in this appendix. [G1] The mapping is an architectural proposal, not a claim that equivalent TOM services already exist. Product availability and naming can change over time.

Consumer operating systems and devices

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Android	TOM Client	Open, portable client runtime with explicit app permissions and provider choice.
Android Auto	TOM Mobility Client	Vehicle interface using portable routes, communications and permissions.
Android TV	TOM Media Client	Open media client with selectable catalogues and recommendation profiles.
Cars with Google built-in	TOM Vehicle Node	Manufacturer-independent map, voice and service interfaces.
Chrome	TOM Browser	Standards-first browser with local vault, query receipts and agent permissions.
Chromebook / ChromeOS	TOM Client OS	Reproducible, portable workspace client rather than mandatory cloud identity.

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Pixel phones, tablet, watch and buds	TOM Reference Devices	Optional open reference hardware; no device line is required by the protocol. Portable health, identity and notification services with local-first data. Multilingual input and translation that records model/provider and stays replaceable. Local and remote file access over user-controlled stores. User-authorized device and product-location graph. User-defined attention budgets and transparent notification rules.
Wear OS	TOM Wearable Client	
Gboard	TOM Input	
Files	TOM Vault Client	
Find Hub	TOM Asset Finder	
Digital Wellbeing	TOM Attention Profile	

Search, AI and knowledge

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Google Search	TOM Query	Federated indexes, evidence bundles, visible ranking profiles and sponsorship separation.
AI Mode / Circle to Search / Lens	TOM Multimodal Query	Image, screen and natural-language queries interpreted as editable need records.
Gemini	TOM Agent	Selectable AI proposal layer under explicit data, tool and action permissions.
Google Assistant	TOM Agent Voice	Voice interface to the same permissioned service graph.
Gemini Notebook / NotebookLM	TOM Research Notebook	Source-bounded synthesis with claim receipts and portable project data.
Scholar	TOM Scholar	Open research graph with datasets, methods, replication and contradiction lineage.
Google Trends	TOM Public Trends	Transparent aggregate query and event statistics with privacy controls.
Finance	TOM Market Data	Multiple attributed financial-data providers and explicit update times.
Translate	TOM Translate	Plural human/model translation with source text, alternatives and terminology packs.
Arts & Culture	TOM Culture Commons	Institution- and community-governed cultural collections with provenance.

Communication, identity and time

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Gmail	TOM Link Mail	Standards-based mail with portable identity, contacts, history and provider migration.
Google Chat	TOM Link Rooms	Federated message rooms with community governance and export.
Google Meet	TOM Link Meetings	Interoperable meetings linked to portable rooms, calendars and decisions.
Google Messages	TOM Link Messaging	Cross-provider messaging with explicit encryption and retention profiles.
Google Voice	TOM Link Voice	Portable calling identity and provider-independent communication records.
Contacts	TOM Relationship Vault	User-owned relationship graph with selective sharing.
Google Calendar	TOM Time	Portable calendar, availability, negotiation and event lineage.
Google Tasks	TOM Work Graph	Typed goals, tasks, dependencies and verified outcomes.

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Google Keep	TOM Notes	Local-first notes linked to projects and evidence.
Google Authenticator	TOM Auth	Open standards authentication and recovery; passkeys/WebAuthn preferred where suitable.
Google Account / Identity Platform	TOM Identity Federation	Plural identifiers and verifiable credentials, no mandatory single global account.

Workspace and storage

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Google Drive	TOM Vault Storage	User-controlled files and provider-portable storage.
Google Docs	TOM Document	Open collaborative document objects with edit and publication lineage.
Google Sheets	TOM Table	Open structured tables with formula, data-source and decision lineage.
Google Slides	TOM Presentation	Portable presentation documents with cited media and version history.
Google Forms	TOM Intake	Typed consent-aware data collection and validation.
Google Sites	TOM Publish	Portable websites and service pages hosted by competing operators.
Google Workspace	TOM Workspace	Unified projects, documents, communication and time over open interfaces.
Google One	TOM Membership + Storage	Cooperative or commercial storage/support plans without ecosystem captivity.
Google Fonts	TOM Design Commons	Open design assets with licence and provenance records.

Maps, mobility and travel

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Google Maps	TOM Atlas	Open geographic graph with source, recency, accessibility and selectable routing.
Google Earth	TOM Earth	Public geospatial, environmental and historical layers with provenance.
Street View	TOM Street Evidence	Attributed street imagery with capture time, privacy and community correction.
Waze	TOM Live Mobility	Federated traffic and hazard observations with confidence and expiry.
Flights	TOM Travel Search	Schedule, price, accessibility, reliability and fee comparison from multiple providers.
Travel	TOM Journey	Portable itinerary and booking plan under explicit permissions.
Google Maps Platform	TOM Atlas APIs	Open map and routing interfaces backed by plural data/operators.

Media, entertainment and publishing

Present Google product or surface	Proposed TOM Commons module	Replacement principle
YouTube	TOM Video Commons	Portable creator identity, hosting competition and user-selected recommendation profiles.

Present Google product or surface	Proposed TOM Commons module	Replacement principle
YouTube Kids	TOM Youth Media	Age-appropriate catalogues with guardian/community governance and no behavioural ad default.
YouTube Music	TOM Music Commons	Portable libraries, artist support, catalogue federation and transparent recommendations.
YouTube TV	TOM Live Media	Federated channel and event discovery with rights/licence metadata.
Google TV	TOM Media Guide	Cross-provider catalogue and device-neutral playback routing.
Google Play Books	TOM Books	Portable purchases, loans, public-library integration and open annotations.
Google Play Games	TOM Games	Portable identity, saves, achievements and community governance.
Google Photos	TOM Memory Vault	User-owned media with local/permissioned AI organisation.
Google News	TOM News Graph	Event, source, claim, correction and ownership graph with selectable editorial profiles.
Blogger	TOM Publish	Federated long-form publishing with portable audience and archive.
Google Cast	TOM Device Media	Open local device-discovery and media-control interfaces.

Education, family, home and wellbeing

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Google Classroom	TOM Learn Classroom	Portable courses, assignments, evidence and credentials across institutions.
Learning / Grow with Google	TOM Learn Commons	Open learning paths linked to demonstrated competence.
Family Link	TOM Family Permissions	Child-sensitive delegated permissions and transparent controls.
Fitbit / Google Fit	TOM Personal Health Vault	User-controlled activity observations; care claims remain clinically governed.
Google Home / Nest	TOM Home	Portable device capabilities, local automation, firmware and repair lineage.
Accessibility support	TOM Access Layer	Accessibility is a cross-cutting conformance requirement, not a separate afterthought.
Crisis Response	TOM Crisis	Public-authority and community emergency data with low-bandwidth/offline modes.

Market, payments and business

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Google Pay	TOM Pay Broker	External regulated payment services under explicit permission and receipt.
Google Wallet	TOM Wallet	Portable credentials, tickets, receipts, warranties and payment references.
Shopping	TOM Market	Product comparison including cost, provenance, reparability, compatibility and sponsorship status.
Business Profile	TOM Service Profile	Portable verified business/service identity independent of one map operator.
Merchant Center	TOM Supplier Registry	Open product, stock, price and policy feeds with seller identity.
Manufacturer Center	TOM Product Passport Registry	Design, manufacturing, certification, repair and recall records.
Local Services Ads	TOM Local Sponsorship	Clearly separated paid placement that cannot rewrite organic service eligibility.

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Google Ads / Search Ads / Shopping Ads	TOM Disclosed Sponsorship	Optional labelled promotion with public rules and no covert ranking mutation.
AdSense / AdMob / Ad Manager	TOM Publisher Funding Exchange	Transparent sponsorship/subscription/membership options controlled by publishers and communities.
Google Analytics / Tag Manager	TOM Outcome Analytics	Purpose-limited service quality and outcome metrics instead of cross-service behavioural enclosure.
Marketing Platform / Demand Gen / Performance Max	TOM Campaign Exchange	Commercial campaign tools separated from public search and personal data by explicit consent.

Cloud and developer platforms

Present Google product or surface	Proposed TOM Commons module	Replacement principle
Google Cloud / Cloud Computing	TOM Compute Federation	Portable reproducible workloads across public, cooperative and commercial operators.
Firebase	TOM App Services	Replaceable identity, data, notifications and serverless services with export contracts.
Flutter	TOM Client Toolkit	Cross-platform open client toolkit; not required by the protocol.
TensorFlow	TOM Model Adapter	External model runtime whose outputs remain typed proposals/evidence.
AI for Developers	TOM Agent SDK	Model and tool adapters with permissions, receipts and tests.
Google for Developers	TOM Builder Commons	Open SDKs, schemas, conformance suites and example services.
Search Console	TOM Publisher Index Console	Transparent indexing, provenance, errors and appeals across index operators.
Android Enterprise / Chrome Enterprise	TOM Organisation Client	Portable device, application and policy management with auditable rules.
Google Play / Play Protect / Play Pass	TOM App Commons	Open signed package catalogues, reproducible builds, safety evidence and portable purchases.
Google Identity Platform	TOM Identity APIs	Interoperable identity and credential verification without one account authority.
Interactive Media Ads	TOM Media Sponsorship	Clearly labelled media sponsorship handled outside organic recommendation.

Appendix B - Human Needs Genome taxonomy

The following taxonomy is intended as a first versioned registry. It is deliberately broad so that no single ministry, corporation or app category owns a life event.

Domain ID	Human need	Representative records and services
HN.01	Air and environmental quality	air observations, alerts, exposure guidance, emissions, indoor systems
HN.02	Water	drinking water, sanitation, supply, quality, flood and drought information
HN.03	Food	availability, affordability, nutrition, production, allergies, cultural needs
HN.04	Shelter	housing, tenancy, ownership, accessibility, energy, maintenance, emergency shelter
HN.05	Health	care navigation, records, prevention, disability, mental health, medicines information
HN.06	Physical safety	emergency response, workplace safety, product safety, violence support

Domain ID	Human need	Representative records and services
HN.07	Identity	identifiers, credentials, pseudonyms, guardianship, recovery and delegation
HN.08	Privacy	consent, selective disclosure, retention, correction, deletion and audit
HN.09	Communication	mail, messaging, calls, meetings, public conversation and translation
HN.10	Relationships	family, friendship, care, community, association and mutual aid
HN.11	Mobility	walking, cycling, public transport, driving, travel, accessibility and logistics
HN.12	Time	calendars, commitments, deadlines, routines, care and resource scheduling
HN.13	Money	payments, income, budgeting, benefits, credit, tax, insurance and receipts
HN.14	Work	jobs, projects, skills, conditions, cooperatives, labour rights and livelihood
HN.15	Learning	curricula, prerequisites, tutoring, assessment, credentials and lifelong learning
HN.16	Knowledge	search, evidence, libraries, archives, research, claims and contradictions
HN.17	Creativity	writing, art, music, design, performance, tools, publication and attribution
HN.18	Culture	language, heritage, religion, tradition, media, local memory and access
HN.19	Recreation	sport, games, travel, events, nature, social activities and rest
HN.20	Home	household assets, devices, energy, maintenance, security and automation
HN.21	Family and care	childcare, education, elder care, delegated decisions and shared records
HN.22	Legal capacity	rights, obligations, documents, representation, dispute and appeal
HN.23	Civic participation	elections, consultation, public meetings, budgets, petitions and representation
HN.24	Public services	eligibility, applications, case status, reasons, correction and redress
HN.25	Justice	legal information, due process, evidence, decisions and rehabilitation
HN.26	Community resilience	local services, mutual aid, emergency plans, public assets and volunteers
HN.27	Infrastructure	transport, energy, water, communications, maintenance and investment
HN.28	Environment	ecosystems, climate, biodiversity, pollution, conservation and adaptation
HN.29	Science	hypotheses, methods, datasets, experiments, replication and open problems
HN.30	Software	source, builds, dependencies, tests, deployment, security and maintenance
HN.31	Manufacturing	requirements, design, materials, process, machines, inspection and product lineage
HN.32	Repair and circularity	diagnostics, parts, skills, warranties, reuse, remanufacture and recycling
HN.33	Biological research	protocols, biosafety, organisms, tissues, measurements and research lineage
HN.34	Functional organs	research, consent, manufacturing evidence, regulation and outcomes
HN.35	Habitat	buildings, settlements, resilience, accessibility and living infrastructure
HN.36	Space and long-term stewardship	missions, off-world manufacturing, archives, governance and intergenerational continuity

Appendix C - Standards and open-data adapter matrix

Foundation	Existing role	TOM Commons use	Boundary
W3C Verifiable Credentials 2.0	Issuer-holder-verifier model for tamper-evident claims	Qualifications, licences, membership, eligibility and selective presentations	Does not define all trust or governance policy
W3C Decentralized Identifiers	Identifiers for people, organisations, physical and digital things	Portable identities for people, services, products, sensors and datasets	Does not by itself create a trusted issuer or consensus system
WebAuthn / passkeys	Public-key authentication	Phishing-resistant user and operator login	Recovery and device loss still need governance
Solid Pods	User-controlled online data stores and interoperable app access	Personal vault and provider migration model	TOM must add domain schemas, evidence and action authority
Matrix	Open federated real-time communication APIs	Messages, rooms, meetings, IoT events and bridgeable communication	Moderation, abuse operations and usability remain institutional work
ActivityPub	Federated actors, inboxes, outboxes and social activities	Publishing, following and community media federation	Does not supply TOM evidence or ranking semantics
OpenStreetMap	Community-maintained open map data	Geographic base, local services and community correction	Routing, service quality and real-time data need additional operators
Wikidata	Collaborative multilingual structured knowledge	Entity and claim graph with source links	Quality and completeness vary by domain
Common Crawl	Open web-crawl corpus	Seed data for public-interest web indexes	A high-quality search engine still needs crawling, parsing, ranking and anti-spam work
OpenAlex	Open research graph	Papers, authors, institutions, topics and citations	Reproducibility and claim-level evidence need additional layers
FHIR	Healthcare data exchange standard	Adapter for patient-controlled and institutional health records	Clinical authority, privacy and regulation remain external
DICOM	Medical imaging standard	Imaging identity and provenance adapter	Interpretation remains clinical/model evidence
OPC UA	Industrial interoperability and information models	Factory, machine and sensor adapters	Physical safety remains with certified industrial controls
FMI	Model exchange and co-simulation	Digital-twin and engineering simulation adapter	Solver correctness remains independently validated
ROS 2 / Matrix/other messaging	Robot and distributed component communication	Non-authoritative sensor/action transport	Does not replace certified motion and emergency controls
PROV-O	General provenance vocabulary	Exchange provenance with external systems	TOM retains its own canonical authority records

Appendix D - Sample proof-carrying query receipts

D.1 Public information query

```
{
  "type": "tom.answer.bundle",
  "need_id": "sha256:...",
  "question": "What public transport is accessible to this address after 21:00?",
  "context_fields_used": ["approximate_origin", "wheelchair_access"],
  "indexes": ["osm-regional-v42", "transit-authority-feed-v8"],
  "ranking_profile": "accessible-reliable-low-transfer-v3",
  "eligible_routes": 4,
  "selected_routes": 2,
  "known_conflicts": ["one lift status feed is stale"],
  "ai_summary": {
    "status": "proposal",
    "model": "selected-provider/model-version",
    "source_claims": ["claim:route-a", "claim:route-b"]
  },
  "action_authority": "human-confirmation-required",
  "receipt_hash": "sha256:..."
}
```

D.2 Public-service application

```
{
  "type": "tom.service.application.plan",
  "need": "income_support",
  "jurisdiction": "declared-local-authority",
  "rule_version": "benefit-rule-2026-09",
  "facts_requested": ["residency", "household_income", "dependants"],
  "possible_eligibility": "requires_authority_decision",
  "missing_evidence": ["latest_income_statement"],
  "submission_steps": 3,
  "appeal_route": "service:appeals-office",
  "data_retention": "jurisdiction-policy-id",
  "ai_role": "plain-language-explanation-only"
}
```

D.3 Manufacturing query

```
{
  "type": "tom.manufacturing.candidate",
  "need": "replacement_appliance_bracket",
  "product_passport": "product:...",
  "part_definition": "definition:...",
  "candidate_fabricator": "service:...",
  "material_compatibility": "pass",
  "machine_capability": "pass",
  "inspection_plan": "inspection:...",
  "simulation": {"status": "external-evidence", "result": "pass"},
  "physical_order": "not-authorised",
  "required_confirmation": ["price", "liability", "fabrication"],
  "lineage_parent": "product:original-appliance"
}
```

Appendix E - Institutional transition map

Present institution or platform role	TOM Commons transition	Human role preserved
Search company	Becomes one of several index/ranking operators	Editors, engineers, librarians, researchers and local contributors
Email provider	Becomes a replaceable communication host	Support, security, abuse response and community administration
Social platform	Becomes federated hosting and recommendation services	Creators, moderators, communities and journalists
Cloud provider	Becomes a portable workload operator	Infrastructure engineering, support and specialist services
Advertising network	Becomes a visibly separated sponsorship exchange	Marketing can continue without controlling organic public search
Government portal	Becomes a standards-compatible public service node	Accountable officials and case workers remain responsible
School platform	Becomes a learning and credential provider	Teachers, institutions and learners retain pedagogical authority
Hospital portal	Becomes a regulated health-data and care service node	Clinicians and institutions retain clinical accountability
Bank/payment app	Becomes a regulated payment/credential provider	Financial institutions retain legal obligations and risk controls
Marketplace	Becomes an open supplier and transaction service	Sellers, buyers, cooperatives and dispute resolvers
Manufacturer	Publishes capabilities and proof-carrying products	Engineers, operators, inspectors and safety professionals
Research publisher	Becomes one node in an open evidence graph	Peer review, editorial judgement and scholarly communities
AI company	Supplies models and agents under explicit scopes	Researchers, reviewers, users and accountable service owners

Present institution or platform role	TOM Commons transition	Human role preserved
Library	Operates public query, archive, literacy and local data services	Librarians become core stewards of the commons
Municipality	Operates local services, maps, notices and appeals	Democratic institutions remain accountable to residents
Standards body	Publishes schemas, conformance tests and revisions	Expert consensus and public review remain essential

Appendix F - The first twelve deliverables

A serious programme should produce concrete artifacts in this order:

1. TOM-HUMAN-NEEDS-1.0.schema.json - typed need, rights, service, evidence, ranking, action and outcome records.
2. tom-vault - encrypted local personal data store with export and provider migration.
3. tom-query-receipt - deterministic record of indexes, filters, ranking, AI use and sources.
4. tom-ranking-public-interest-v1 - first inspectable ranking profile.
5. tom-open-research-index - bounded OpenAlex/Wikidata research explorer.
6. tom-local-atlas - one-region OpenStreetMap service and accessibility directory.
7. tom-link-bridge - Matrix/email bridge with portable contacts and rooms.
8. tom-service-registry - one municipality or community's machine-readable services.
9. tom-agent-gate - local/API model proposals with source receipts and action confirmation.
10. tom-repair-foundry-demo - need-to-proof-carrying replacement part workflow.
11. tom-commons-conformance - interoperability, portability, security and accessibility tests.
12. tom-commons-cooperative-constitution - governance, funding, audit, appeals and forkability.

References

Supplied TOM materials

- **[T1]** TOM_seed_genome_2026-09-01.txt. Exact 244-byte canonical seed, supplied by the requester.
- **[T2]** TOMAGI: *Topological Operator Machine for Analytic Geometric Inference*, version 1.0.0, 1 September 2026. Supplied project specification.
- **[T3]** TOM *Topological Open Modular Seeded Referential Substrate*, TOM-SRS 1.0. Supplied project specification.
- **[T4]** CODEX_KERNEL_0_5_2_REPAIR_HANDOFF.md and proof JSON. Supplied corrective authority boundary.
- **[T5]** TOM *Manufacturing Horizons: From a Frozen Deterministic Kernel to Frontier Physical, Biochemical and Biological Fabrication*, 3 September 2026. Companion source document.

Current platform and financial sources

- **[G1]** Google, "Google's products and services - About Google." Product directory accessed 3 September 2026. <https://about.google/products/>
- **[G2]** Alphabet Inc., Annual Report on Form 10-K for the year ended 31 December 2025, filed 2026. Product and revenue descriptions, advertising dependence and disaggregated revenues. <https://www.sec.gov/Archives/edgar/data/1652044/000165204426000018/goog-20251231.htm>

Open protocols and commons foundations

- **[W1]** World Wide Web Consortium, *Verifiable Credentials Data Model v2.0*, W3C Recommendation, 15 May 2025. <https://www.w3.org/TR/vc-data-model-2.0/>
- **[W2]** World Wide Web Consortium, *Decentralized Identifiers (DIDs) v1.0*. <https://www.w3.org/TR/did/>
- **[W3]** World Wide Web Consortium, *ActivityPub*. <https://www.w3.org/TR/activitypub/>
- **[W4]** Solid Project, "About Solid." <https://solidproject.org/about>
- **[W5]** Matrix.org Foundation, *Matrix Specification*. <https://spec.matrix.org/latest/>

- **[W6]** OpenStreetMap, “About OpenStreetMap.” <https://www.openstreetmap.org/about>
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- **[W8]** Common Crawl, “Overview.” <https://commoncrawl.org/overview>
- **[W9]** OpenAlex, open research graph. <https://openalex.org/>
- **[W10]** World Wide Web Consortium, *Web Authentication: An API for accessing Public Key Credentials*. <https://www.w3.org/TR/webauthn-3/>

Human rights and digital public infrastructure

- **[H1]** United Nations, *Universal Declaration of Human Rights*. <https://www.un.org/en/about-us/universal-declaration-of-human-rights>
- **[H2]** United Nations Development Programme, “Digital public infrastructure.” <https://www.undp.org/digital-public-infrastructure>

Appendix G - Closing note

TOM Commons is intentionally described as a replacement architecture rather than a finished brand. The word “Google” in this report names the breadth of functions to be replaced: discovery, interpretation, identity, communication, work, maps, media, commerce, cloud, AI and the route from information to action. A credible successor cannot be built by copying one interface while retaining the same centralised control structure.

The evolutionary move is from:

one company account
 one hidden ranking economy
 one behavioural profile
 one integrated service enclosure

To:

portable human context
 plural indexes and models
 visible evidence and ranking
 interoperable services
 permissioned action
 proof-carrying outcomes
 right to exit and appeal

That transition can begin with ordinary computers and open protocols. The most futuristic manufacturing and biological horizons remain downstream research programmes. The fixed TOM substrate contributes deterministic identity, bounded authority, replay and lineage; society still has to build the institutions, science, care and democratic judgement that make those capabilities humane.